Gradient descent without step size calculation: divergent

```python
import numpy as np

def f(x):
    return x[0]**2 + 10 * x[1]**2

def gradf(x):
    return np.array([2 * x[0], 20 * x[1]])

xk = np.array([2.0,2.0])
epsilon = 0.001
print(0,xk,f(xk))

for k in range(0,120):
    dk = -gradf(xk)
    if np.linalg.norm(dk) < epsilon:
        break

    xk = xk + dk
    print('k',k+1,'xk',xk,'f(xk)',f(xk))

print('xk',xk,'f(xk)',f(xk))
```

```
0 [2. 2.] 44.0
k 1 xk [ -2. -38.] f(xk) 14444.0
k 2 xk [  2. 722.] f(xk) 5212844.0
k 3 xk [-2.0000e+00 -1.3718e+04] f(xk) 1881835244.0
k 4 xk [2.00000e+00 2.60642e+05] f(xk) 679342521644.0
k 5 xk [-2.000000e+00 -4.952198e+06] f(xk) 245242650312044.0
k 6 xk [2.0000000e+00 9.4091762e+07] f(xk) 8.853259676264643e+16
k 7 xk [-2.00000000e+00 -1.78774348e+09] f(xk) 3.1960267431315366e+19
k 8 xk [2.00000000e+00 3.39671261e+10] f(xk) 1.1537656542704846e+22
k 9 xk [-2.00000000e+00 -6.45375396e+11] f(xk) 4.1650940119164497e+24
k 10 xk [2.00000000e+00 1.22621325e+13] f(xk) 1.5035989383018384e+27
k 11 xk [-2.00000000e+00 -2.32980518e+14] f(xk) 5.427992167269636e+29
k 12 xk [2.00000000e+00 4.42662984e+15] f(xk) 1.9595051723843386e+32
k 13 xk [-2.00000000e+00 -8.41059669e+16] f(xk) 7.073813672307462e+34
k 14 xk [2.00000000e+00 1.59801337e+18] f(xk) 2.553646735702994e+37
k 15 xk [-2.00000000e+00 -3.03622541e+19] f(xk) 9.218664715887809e+39
k 16 xk [2.00000000e+00 5.76882827e+20] f(xk) 3.3279379624354996e+42
k 17 xk [-2.00000000e+00 -1.09607737e+22] f(xk) 1.2013856044392152e+45
k 18 xk [2.00000000e+00 2.08254701e+23] f(xk) 4.337002032025568e+47
k 19 xk [-2.00000000e+00 -3.95683931e+24] f(xk) 1.56565773356123e+50
k 20 xk [2.00000000e+00 7.51799469e+25] f(xk) 5.65202441815604e+52
k 21 xk [-2.00000000e+00 -1.42841899e+27] f(xk) 2.040380814954331e+55
k 22 xk [2.00000000e+00 2.71399608e+28] f(xk) 7.365774741985135e+57
k 23 xk [-2.00000000e+00 -5.15659256e+29] f(xk) 2.659044681856634e+60
k 24 xk [2.00000000e+00 9.79752586e+30] f(xk) 9.599151301502448e+62
```

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k 25 xk [-2.00000000e+00 -1.86152991e+32] f(xk) 3.465293619842383e+65
k 26 xk [2.00000000e+00 3.53690684e+33] f(xk) 1.2509709967631004e+68
k 27 xk [-2.00000000e+00 -6.72012299e+34] f(xk) 4.516005298314793e+70
k 28 xk [2.00000000e+00 1.27682337e+36] f(xk) 1.6302779126916407e+73
k 29 xk [-2.0000000e+00 -2.4259644e+37] f(xk) 5.885303264816822e+75
k 30 xk [2.00000000e+00 4.60933236e+38] f(xk) 2.1245944785988723e+78
k 31 xk [-2.00000000e+00 -8.75773148e+39] f(xk) 7.669786067741929e+80
k 32 xk [2.00000000e+00 1.66396898e+41] f(xk) 2.7687927704548367e+83
k 33 xk [-2.00000000e+00 -3.16154106e+42] f(xk) 9.995341901341963e+85
k 34 xk [2.00000000e+00 6.00692802e+43] f(xk) 3.6083184263844476e+88
k 35 xk [-2.00000000e+00 -1.14131632e+45] f(xk) 1.3026029519247853e+91
k 36 xk [2.00000000e+00 2.16850102e+46] f(xk) 4.702396656448474e+93
k 37 xk [-2.00000000e+00 -4.12015193e+47] f(xk) 1.6975651929778992e+96
k 38 xk [2.00000000e+00 7.82828867e+48] f(xk) 6.128210346650217e+98
k 39 xk [-2.00000000e+00 -1.48737485e+50] f(xk)
2.2122839351407283e+101
k 40 xk [2.00000000e+00 2.82601221e+51] f(xk) 7.986345005858028e+103
k 41 xk [-2.0000000e+00 -5.3694232e+52] f(xk) 2.883070547114749e+106
k 42 xk [2.00000000e+00 1.02019041e+54] f(xk) 1.0407884675084245e+109
k 43 xk [-2.00000000e+00 -1.93836177e+55] f(xk)
3.7572463677054123e+111
k 44 xk [2.00000000e+00 3.68288737e+56] f(xk) 1.3563659387416538e+114
k 45 xk [-2.000000e+00 -6.997486e+57] f(xk) 4.896481038857371e+116
k 46 xk [2.00000000e+00 1.32952234e+59] f(xk) 1.767629655027511e+119
k 47 xk [-2.00000000e+00 -2.52609245e+60] f(xk) 6.381143054649315e+121
k 48 xk [2.00000000e+00 4.79957565e+61] f(xk) 2.3035926427284025e+124
k 49 xk [-2.00000000e+00 -9.11919374e+62] f(xk) 8.315969440249534e+126
k 50 xk [2.00000000e+00 1.73264681e+64] f(xk) 3.002064967930082e+129
k 51 xk [-2.00000000e+00 -3.29202894e+65] f(xk)
1.0837454534227597e+132
k 52 xk [2.00000000e+00 6.25485498e+66] f(xk) 3.912321086856163e+134
k 53 xk [-2.00000000e+00 -1.18842245e+68] f(xk)
1.4123479123550747e+137
k 54 xk [2.00000000e+00 2.25800265e+69] f(xk) 5.098575963601821e+139
k 55 xk [-2.00000000e+00 -4.29020503e+70] f(xk) 1.840585922860257e+142
k 56 xk [2.00000000e+00 8.15138956e+71] f(xk) 6.644515181525528e+144
k 57 xk [-2.00000000e+00 -1.54876402e+73] f(xk) 2.398669980530716e+147
k 58 xk [2.00000000e+00 2.94265163e+74] f(xk) 8.659198629715889e+149
k 59 xk [-2.0000000e+00 -5.5910381e+75] f(xk) 3.1259707053274363e+152
k 60 xk [2.00000000e+00 1.06229724e+77] f(xk) 1.1284754246232043e+155
k 61 xk [-2.00000000e+00 -2.01836475e+78] f(xk) 4.073796282889768e+157
k 62 xk [2.00000000e+00 3.83489303e+79] f(xk) 1.4706404581232057e+160
k 63 xk [-2.00000000e+00 -7.28629676e+80] f(xk) 5.309012053824772e+162
k 64 xk [2.00000000e+00 1.38439639e+82] f(xk) 1.916553351430743e+165
k 65 xk [-2.00000000e+00 -2.63035313e+83] f(xk) 6.918757598664983e+167
k 66 xk [2.00000000e+00 4.99767095e+84] f(xk) 2.497671493118058e+170
k 67 xk [-2.00000000e+00 -9.49557481e+85] f(xk) 9.016594090156187e+172
k 68 xk [2.00000000e+00 1.80415921e+87] f(xk) 3.2549904665463845e+175
k 69 xk [-2.00000000e+00 -3.42790251e+88] f(xk)
```

```
1.1750515584232445e+178
k 70 xk [2.00000000e+00 6.51301476e+89] f(xk) 4.241936125907914e+180
k 71 xk [-2.0000000e+00 -1.2374728e+91] f(xk) 1.531338941452757e+183
k 72 xk [2.00000000e+00 2.35119833e+92] f(xk) 5.528133578644453e+185
k 73 xk [-2.00000000e+00 -4.46727682e+93] f(xk) 1.995656221890647e+188
k 74 xk [2.00000000e+00 8.48782596e+94] f(xk) 7.204318961025237e+190
k 75 xk [-2.00000000e+00 -1.61268693e+96] f(xk)
2.6007591449301106e+193
k 76 xk [2.00000000e+00 3.06410517e+97] f(xk) 9.388740513197702e+195
k 77 xk [-2.00000000e+00 -5.82179983e+98] f(xk)
3.3893353252643693e+198
k 78 xk [2.00000000e+000 1.10614197e+100] f(xk)
1.2235500524204378e+201
k 79 xk [-2.00000000e+000 -2.10166974e+101] f(xk)
4.417015689237779e+203
k 80 xk [2.0000000e+000 3.9931725e+102] f(xk) 1.594542663814838e+206
k 81 xk [-2.00000000e+000 -7.58702776e+103] f(xk)
5.756299016371566e+208
k 82 xk [2.00000000e+000 1.44153527e+105] f(xk)
2.0780239449101352e+211
k 83 xk [-2.00000000e+000 -2.73891702e+106] f(xk)
7.501666441125588e+213
k 84 xk [2.00000000e+000 5.20394234e+107] f(xk)
2.7081015852463372e+216
k 85 xk [-2.00000000e+000 -9.88749044e+108] f(xk)
9.776246722739277e+218
k 86 xk [2.00000000e+000 1.87862318e+110] f(xk)
3.5292250669088776e+221
k 87 xk [-2.00000000e+000 -3.56938405e+111] f(xk)
1.2740502491541045e+224
k 88 xk [2.00000000e+000 6.78182969e+112] f(xk)
4.5993213994463185e+226
k 89 xk [-2.00000000e+000 -1.28854764e+114] f(xk)
1.6603550252001206e+229
k 90 xk [2.00000000e+000 2.44824052e+115] f(xk)
5.9938816409724346e+231
k 91 xk [-2.00000000e+000 -4.65165699e+116] f(xk)
2.1637912723910492e+234
k 92 xk [2.00000000e+000 8.83814828e+117] f(xk) 7.811286493331688e+236
k 93 xk [-2.00000000e+000 -1.67924817e+119] f(xk)
2.8198744240927387e+239
k 94 xk [2.00000000e+000 3.19057153e+120] f(xk)
1.0179746670974788e+242
k 95 xk [-2.0000000e+000 -6.0620859e+121] f(xk)
3.6748885482218987e+244
k 96 xk [2.00000000e+000 1.15179632e+123] f(xk) 1.326634765908106e+247
k 97 xk [-2.00000000e+000 -2.18841301e+124] f(xk)
4.789151504928263e+249
k 98 xk [2.00000000e+000 4.15798472e+125] f(xk)
```

```
1.7288836932791027e+252
k 99 xk [-2.00000000e+000 -7.90017097e+126] f(xk)
6.241270132737561e+254
k 100 xk [2.00000000e+000 1.50103248e+128] f(xk)
2.2530985179182595e+257
k 101 xk [-2.00000000e+000 -2.85196172e+129] f(xk)
8.133685649684918e+259
k 102 xk [2.00000000e+000 5.41872727e+130] f(xk)
2.936260519536255e+262
k 103 xk [-2.00000000e+000 -1.02955818e+132] f(xk)
1.0599900475525882e+265
k 104 xk [2.00000000e+000 1.95616054e+133] f(xk)
3.8265640716648437e+267
k 105 xk [-2.00000000e+000 -3.71670503e+134] f(xk)
1.3813896298710086e+270
k 106 xk [2.00000000e+000 7.06173956e+135] f(xk)
4.986816563834341e+272
k 107 xk [-2.00000000e+000 -1.34173052e+137] f(xk)
1.8002407795441973e+275
k 108 xk [2.00000000e+000 2.54928798e+138] f(xk)
6.498869214154553e+277
k 109 xk [-2.00000000e+000 -4.84364717e+139] f(xk)
2.346091786309794e+280
k 110 xk [2.00000000e+000 9.20292961e+140] f(xk)
8.469391348578356e+282
k 111 xk [-2.00000000e+000 -1.74855663e+142] f(xk)
3.057450276836786e+285
k 112 xk [2.00000000e+000 3.32225759e+143] f(xk)
1.1037395499380795e+288
k 113 xk [-2.00000000e+000 -6.31228942e+144] f(xk)
3.984499775276468e+290
k 114 xk [2.00000000e+000 1.19933499e+146] f(xk)
1.4384044188748048e+293
k 115 xk [-2.00000000e+000 -2.27873648e+147] f(xk)
5.192639952138044e+295
k 116 xk [2.00000000e+000 4.32959931e+148] f(xk)
1.8745430227218336e+298
k 117 xk [-2.0000000e+000 -8.2262387e+149] f(xk)
6.767100312025818e+300
k 118 xk [2.00000000e+000 1.56298535e+151] f(xk)
2.442923212641321e+303
k 119 xk [-2.00000000e+000 -2.96967217e+152] f(xk)
8.818952797635167e+305
k 120 xk [2.00000000e+000 5.64237712e+153] f(xk) inf
xk [2.00000000e+000 5.64237712e+153] f(xk) inf

C:\Users\winter\AppData\Local\Temp\ipykernel_6692\1328853885.py:4:
RuntimeWarning: overflow encountered in scalar multiply
  return x[0]**2 + 10 * x[1]**2
```

Gradient descent with Armijo-Goldstein step size calculation: convergent but slow

```
xk = np.array([2.0,2.0])
epsilon = 0.00001
beta = 0.9
sigma = 0.01

print(0,xk,f(xk))

for k in range(0,20000):
    dk = -gradf(xk)
    if np.linalg.norm(dk) < epsilon:
        break

    def phi(c):
        return f(xk + c * dk)

    def dphi(c):
        return np.dot(gradf(xk + c * dk),dk)

    ck = 1.0

    while phi(ck) > phi(0) + sigma * ck * dphi(0):
        #print('ck',ck,'phi(ck)',phi(ck),'target:',phi(0) + sigma * ck
* dphi(0))
        ck = ck * beta

    #print('ck',ck,'phi(ck)',phi(ck),'target:',phi(0) + sigma * ck *
dphi(0))
    print('Gradient',dk,'Step size',ck)

    xk = xk + ck * dk
    print('k',k+1,'xk',xk,'f(xk)',f(xk))

print('xk',xk,'f(xk)',f(xk))
```

```
0 [2. 2.] 44.0
Gradient [ -4. -40.] Step size 0.0984770902183612
k 1 xk [ 1.60609164 -1.93908361] f(xk) 40.17998276989832
Gradient [-3.21218328 38.78167217] Step size 0.0984770902183612
k 2 xk [1.28976518 1.88002262] f(xk) 37.008344759237254
Gradient [ -2.57953035 -37.60045242] Step size 0.0984770902183612
k 3 xk [ 1.03574053 -1.82276052] f(xk) 34.29731773277605
Gradient [-2.07148107 36.45521048] Step size 0.0984770902183612
k 4 xk [0.83174711 1.76724253] f(xk) 31.923264754566283
Gradient [ -1.66349421 -35.34485055] Step size 0.0984770902183612
k 5 xk [ 0.66793104 -1.71341551] f(xk) 29.804058926339327
Gradient [-1.33586207 34.26831018] Step size 0.0984770902183612
k 6 xk [0.53637923 1.66122796] f(xk) 27.884486158731658
Gradient [ -1.07275845 -33.22455928] Step size 0.0984770902183612
```

```
k 7 xk [ 0.4307371  -1.61062996] f(xk) 26.12682305173391
Gradient [-0.86147419 32.21259915] Step size 0.0984770902183612
k 8 xk [0.34590162 1.56157308] f(xk) 24.504752630125996
Gradient [ -0.69180325 -31.23146151] Step size 0.0984770902183612
k 9 xk [ 0.27777485 -1.51401038] f(xk) 22.999433089606725
Gradient [-0.55554971 30.28020754] Step size 0.0984770902183612
k 10 xk [0.22306593 1.46789635] f(xk) 21.596955439001956
Gradient [ -0.44613187 -29.35792706] Step size 0.0984770902183612
k 11 xk [ 0.17913217 -1.42318688] f(xk) 20.28669724692706
Gradient [-0.35826433 28.46373757] Step size 0.0984770902183612
k 12 xk [0.14385134 1.37983917] f(xk) 19.060254674397
Gradient [ -0.28770267 -27.59678348] Step size 0.0984770902183612
k 13 xk [ 0.11551921 -1.33781176] f(xk) 17.91074781354009
Gradient [-0.23103843 26.75623525] Step size 0.0984770902183612
k 14 xk [0.09276722 1.29706443] f(xk) 16.832367121713656
Gradient [ -0.18553445 -25.94128861] Step size 0.0984770902183612
k 15 xk [ 0.07449633 -1.25755819] f(xk) 15.82007566923807
Gradient [-0.14899266 25.15116376] Step size 0.0984770902183612
k 16 xk [0.05982397 1.21925523] f(xk) 14.86941218389253
Gradient [ -0.11964793 -24.3851047 ] Step size 0.0984770902183612
k 17 xk [ 0.04804139 -1.18211892] f(xk) 13.976359393846495
Gradient [-0.09608277 23.64237841] Step size 0.0984770902183612
k 18 xk [0.03857943 1.14611371] f(xk) 13.137254758996574
Gradient [ -0.07715887 -22.92227422] Step size 0.0984770902183612
k 19 xk [ 0.03098105 -1.11120516] f(xk) 12.348728799717254
Gradient [-0.06196211 22.22410311] Step size 0.0984770902183612
k 20 xk [0.02487921 1.07735985] f(xk) 11.607661468863515
Gradient [ -0.04975841 -21.54719703] Step size 0.0984770902183612
k 21 xk [ 0.01997914 -1.04454541] f(xk) 10.911150390945725
Gradient [-0.03995828 20.89090829] Step size 0.0984770902183612
k 22 xk [0.01604417 1.01273045] f(xk) 10.256486971749643
Gradient [ -0.03208833 -20.25460891] Step size 0.0984770902183612
k 23 xk [ 0.0128842 -0.9818845] f(xk) 9.64113778793352
Gradient [-0.0257684  19.63769007] Step size 0.0984770902183612
k 24 xk [0.0103466  0.95197807] f(xk) 9.062729573789046
Gradient [ -0.02069321 -19.03956147] Step size 0.0984770902183612
k 25 xk [ 0.0083088  -0.92298254] f(xk) 8.519036708419366
Gradient [-1.66175935e-02  1.84596508e+01] Step size
0.0984770902183612
k 26 xk [0.00667234 0.89487016] f(xk) 8.007970485223264
Gradient [-1.33446890e-02 -1.78974031e+01] Step size
0.0984770902183612
k 27 xk [ 0.0053582  -0.86761403] f(xk) 7.527569690400389
Gradient [-1.07163967e-02  1.73522805e+01] Step size
0.0984770902183612
k 28 xk [0.00430288 0.84118807] f(xk) 7.075992175687299
Gradient [-8.60575759e-03 -1.68237614e+01] Step size
0.0984770902183612
k 29 xk [ 0.00345541 -0.815567  ] f(xk) 6.6515072133195074
```

```
Gradient [-6.91081766e-03  1.63113399e+01] Step size
0.0984770902183612
k 30 xk [0.00277485 0.7907263 ] f(xk) 6.2524884880389076
Gradient [-5.54970323e-03 -1.58145260e+01] Step size
0.0984770902183612
k 31 xk [ 0.00222833 -0.7666422 ] f(xk) 5.87740762456766
Gradient [-4.45666598e-03  1.53328440e+01] Step size
0.0984770902183612
k 32 xk [0.00178945 0.74329166] f(xk) 5.524828177564589
Gradient [-3.57890698e-03 -1.48658333e+01] Step size
0.0984770902183612
k 33 xk [ 0.00143701 -0.72065234] f(xk) 5.193400029969202
Gradient [-2.87402629e-03  1.44130468e+01] Step size
0.0984770902183612
k 34 xk [0.00115399 0.69870257] f(xk) 4.881854158241223
Gradient [-2.30797480e-03 -1.39740514e+01] Step size
0.0984770902183612
k 35 xk [ 0.0009267  -0.67742135] f(xk) 4.588997731527374
Gradient [-1.85340951e-03  1.35484270e+01] Step size
0.0984770902183612
k 36 xk [0.00074419 0.65678832] f(xk) 4.313709517656355
Gradient [-1.48837276e-03 -1.31357664e+01] Step size
0.0984770902183612
k 37 xk [ 5.97615762e-04 -6.36783732e-01] f(xk) 4.054935572998389
Gradient [-1.19523152e-03  1.27356746e+01] Step size
0.0984770902183612
k 38 xk [4.79912840e-04 6.17388449e-01] f(xk) 3.8116851962216307
Gradient [-9.59825679e-04 -1.23477690e+01] Step size
0.0984770902183612
k 39 xk [ 3.85392000e-04 -5.98583911e-01] f(xk) 3.5830271282192703
Gradient [-7.70783999e-04  1.19716782e+01] Step size
0.0984770902183612
k 40 xk [3.09487434e-04 5.80352125e-01] f(xk) 3.3680859822176448
Gradient [-6.18974868e-04 -1.16070425e+01] Step size
0.0984770902183612
k 41 xk [ 2.48532590e-04 -5.62675646e-01] f(xk) 3.16603888946884
Gradient [-4.97065181e-04  1.12535129e+01] Step size
0.0984770902183612
k 42 xk [1.99583058e-04 5.45537561e-01] f(xk) 2.9761123470868767
Gradient [-3.99166115e-04 -1.09107512e+01] Step size
0.0984770902183612
k 43 xk [ 1.60274340e-04 -5.28921471e-01] f(xk) 2.7975792555733583
Gradient [-3.20548680e-04  1.05784294e+01] Step size
0.0984770902183612
k 44 xk [1.28707639e-04 5.12811478e-01] f(xk) 2.629756134441931
Gradient [-2.57415278e-04 -1.02562296e+01] Step size
0.0984770902183612
k 45 xk [ 1.03358131e-04 -4.97192166e-01] f(xk) 2.4720005051213594
Gradient [-2.06716263e-04  9.94384331e+00] Step size
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0.0984770902183612
k 46 xk [8.30013152e-05 4.82048589e-01] f(xk) 2.323708431014451
Gradient [-1.66002630e-04 -9.64097179e+00] Step size
0.0984770902183612
k 47 xk [ 6.66538592e-05 -4.67366259e-01] f(xk) 2.1843122052285864
Gradient [-1.33307718e-04  9.34732518e+00] Step size
0.0984770902183612
k 48 xk [5.35261030e-05 4.53131126e-01] f(xk) 2.053278177082647
Gradient [-1.07052206e-04 -9.06262252e+00] Step size
0.0984770902183612
k 49 xk [ 4.29839133e-05 -4.39329570e-01] f(xk) 1.9301047090417638
Gradient [-8.59678265e-05  8.78659139e+00] Step size
0.0984770902183612
k 50 xk [3.45180519e-05 4.25948384e-01] f(xk) 1.814320256240484
Gradient [-6.90361037e-05 -8.51896767e+00] Step size
0.0984770902183612
k 51 xk [ 2.77195772e-05 -4.12974764e-01] f(xk) 1.7054815612306338
Gradient [-5.54391545e-05  8.25949529e+00] Step size
0.0984770902183612
k 52 xk [2.22600906e-05 4.00396298e-01] f(xk) 1.603171957035371
Gradient [-4.45201812e-05 -8.00792597e+00] Step size
0.0984770902183612
k 53 xk [ 1.78758727e-05 -3.88200949e-01] f(xk) 1.5069997720082056
Gradient [-3.57517454e-05  7.76401899e+00] Step size
0.0984770902183612
k 54 xk [1.43551449e-05 3.76377049e-01] f(xk) 1.4165968303872152
Gradient [-2.87102897e-05 -7.52754098e+00] Step size
0.0984770902183612
k 55 xk [ 1.15278391e-05 -3.64913283e-01] f(xk) 1.3316170428021286
Gradient [-2.30556781e-05  7.29826566e+00] Step size
0.0984770902183612
k 56 xk [9.25738297e-06 3.53798683e-01] f(xk) 1.2517350813370245
Gradient [-1.85147659e-05 -7.07597366e+00] Step size
0.0984770902183612
k 57 xk [ 7.43410269e-06 -3.43022614e-01] f(xk) 1.176645134075567
Gradient [-1.48682054e-05  6.86045227e+00] Step size
0.0984770902183612
k 58 xk [5.96992509e-06 3.32574764e-01] f(xk) 1.1060597343602645
Gradient [-1.19398502e-05 -6.65149527e+00] Step size
0.0984770902183612
k 59 xk [ 4.79412339e-06 -3.22445136e-01] f(xk) 1.0397086602834622
Gradient [-9.58824677e-06  6.44890273e+00] Step size
0.0984770902183612
k 60 xk [3.84990074e-06 3.12624039e-01] f(xk) 0.9773379001967759
Gradient [-7.69980149e-06 -6.25248079e+00] Step size
0.0984770902183612
k 61 xk [ 3.09164670e-06 -3.03102075e-01] f(xk) 0.9187086802784787
Gradient [-6.18329340e-06  6.06204151e+00] Step size
0.0984770902183612
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k 62 xk [2.48273396e-06 2.93870133e-01] f(xk) 0.8635965504359865
Gradient [-4.96546791e-06 -5.87740266e+00] Step size
0.0984770902183612
k 63 xk [ 1.99374913e-06 -2.84919379e-01] f(xk) 0.8117905250439499
Gradient [-3.98749825e-06  5.69838758e+00] Step size
0.0984770902183612
k 64 xk [1.60107190e-06 2.76241249e-01] f(xk) 0.7630922752284005
Gradient [-3.20214380e-06 -5.52482498e+00] Step size
0.0984770902183612
k 65 xk [ 1.28573410e-06 -2.67827439e-01] f(xk) 0.7173153696047535
Gradient [-2.57146819e-06  5.35654878e+00] Step size
0.0984770902183612
k 66 xk [1.03250339e-06 2.59669898e-01] f(xk) 0.6742845605629642
Gradient [-2.06500678e-06 -5.19339796e+00] Step size
0.0984770902183612
k 67 xk [ 8.29147532e-07 -2.51760822e-01] f(xk) 0.6338351133675197
Gradient [-1.65829506e-06  5.03521643e+00] Step size
0.0984770902183612
k 68 xk [6.65843459e-07 2.44092641e-01] f(xk) 0.5958121755038527
Gradient [-1.33168692e-06 -4.88185283e+00] Step size
0.0984770902183612
k 69 xk [ 5.34702806e-07 -2.36658020e-01] f(xk) 0.5600701838568463
Gradient [-1.06940561e-06  4.73316040e+00] Step size
0.0984770902183612
k 70 xk [4.29390853e-07 2.29449844e-01] f(xk) 0.5264723074519249
Gradient [-8.58781707e-07 -4.58899687e+00] Step size
0.0984770902183612
k 71 xk [ 3.44820530e-07 -2.22461215e-01] f(xk) 0.4948899236253813
Gradient [-6.89641060e-07  4.44922431e+00] Step size
0.0984770902183612
k 72 xk [2.76906685e-07 2.15685448e-01] f(xk) 0.4652021256185618
Gradient [-5.53813370e-07 -4.31370896e+00] Step size
0.0984770902183612
k 73 xk [ 2.22368756e-07 -2.09116059e-01] f(xk) 0.43729525971082855
Gradient [-4.44737512e-07  4.18232117e+00] Step size
0.0984770902183612
k 74 xk [1.78572300e-07 2.02746761e-01] f(xk) 0.4110624901193142
Gradient [-3.57144599e-07 -4.05493522e+00] Step size
0.0984770902183612
k 75 xk [ 1.43401739e-07 -1.96571460e-01] f(xk) 0.3864033899997683
Gradient [-2.86803478e-07  3.93142921e+00] Step size
0.0984770902183612
k 76 xk [1.15158167e-07 1.90584248e-01] f(xk) 0.3632235569827265
Gradient [-2.30316334e-07 -3.81168497e+00] Step size
0.0984770902183612
k 77 xk [ 9.24772845e-08 -1.84779396e-01] f(xk) 0.3414342517731631
Gradient [-1.84954569e-07  3.69558792e+00] Step size
0.0984770902183612
k 78 xk [7.42634967e-08 1.79151349e-01] f(xk) 0.3209520584300764
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Gradient [-1.48526993e-07 -3.58302698e+00] Step size
0.0984770902183612
k 79 xk [ 5.96369906e-08 -1.73694722e-01] f(xk) 0.30169856502545556
Gradient [-1.19273981e-07  3.47389444e+00] Step size
0.0984770902183612
k 80 xk [4.78912360e-08 1.68404294e-01] f(xk) 0.2836000634600992
Gradient [-9.57824719e-08 -3.36808589e+00] Step size
0.0984770902183612
k 81 xk [ 3.84588568e-08 -1.63275003e-01] f(xk) 0.26658726728708926
Gradient [-7.69177137e-08  3.26550007e+00] Step size
0.0984770902183612
k 82 xk [3.08842242e-08 1.58301941e-01] f(xk) 0.2505950464626641
Gradient [-6.17684484e-08 -3.16603883e+00] Step size
0.0984770902183612
k 83 xk [ 2.48014471e-08 -1.53480350e-01] f(xk) 0.23556217800903984
Gradient [-4.96028943e-08  3.06960700e+00] Step size
0.0984770902183612
k 84 xk [1.99166984e-08 1.48805615e-01] f(xk) 0.22143111163464252
Gradient [-3.98333969e-08 -2.97611231e+00] Step size
0.0984770902183612
k 85 xk [ 1.59940214e-08 -1.44273265e-01] f(xk) 0.2081477494144747
Gradient [-3.19880429e-08  2.88546530e+00] Step size
0.0984770902183612
k 86 xk [1.28439320e-08 1.39878961e-01] f(xk) 0.19566123868716911
Gradient [-2.56878641e-08 -2.79757923e+00] Step size
0.0984770902183612
k 87 xk [ 1.03142659e-08 -1.35618501e-01] f(xk) 0.18392377737587548
Gradient [-2.06285319e-08  2.71237001e+00] Step size
0.0984770902183612
k 88 xk [8.28282814e-09 1.31487806e-01] f(xk) 0.17289043098769336
Gradient [-1.65656563e-08 -2.62975612e+00] Step size
0.0984770902183612
k 89 xk [ 6.65149051e-09 -1.27482925e-01] f(xk) 0.16251896059106852
Gradient [-1.33029810e-08  2.54965849e+00] Step size
0.0984770902183612
k 90 xk [5.34145165e-09 1.23600025e-01] f(xk) 0.15276966111260007
Gradient [-1.06829033e-08 -2.47200049e+00] Step size
0.0984770902183612
k 91 xk [ 4.28943042e-09 -1.19835391e-01] f(xk) 0.14360520933421028
Gradient [-8.57886084e-09  2.39670782e+00] Step size
0.0984770902183612
k 92 xk [3.44460917e-09 1.16185421e-01] f(xk) 0.13499052100876507
Gradient [-6.88921833e-09 -2.32370842e+00] Step size
0.0984770902183612
k 93 xk [ 2.76617899e-09 -1.12646623e-01] f(xk) 0.1268926165471409
Gradient [-5.53235798e-09  2.25293246e+00] Step size
0.0984770902183612
k 94 xk [2.22136847e-09 1.09215610e-01] f(xk) 0.11928049476254882
Gradient [-4.44273695e-09 -2.18431220e+00] Step size
```

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0.0984770902183612
k 95 xk [ 1.78386067e-09 -1.05889100e-01] f(xk) 0.1121250141887708
Gradient [-3.56772133e-09  2.11778199e+00] Step size
0.0984770902183612
k 96 xk [1.43252185e-09 1.02663909e-01] f(xk) 0.10539878151795998
Gradient [-2.86504370e-09 -2.05327817e+00] Step size
0.0984770902183612
k 97 xk [ 1.15038068e-09 -9.95369513e-02] f(xk) 0.09907604673091058
Gradient [-2.30076137e-09  1.99073903e+00] Step size
0.0984770902183612
k 98 xk [9.23808399e-10 9.65052354e-02] f(xk) 0.09313260451832567
Gradient [-1.84761680e-09 -1.93010471e+00] Step size
0.0984770902183612
k 99 xk [ 7.41860473e-10 -9.35658600e-02] f(xk) 0.08754570161569403
Gradient [-1.48372095e-09  1.87131720e+00] Step size
0.0984770902183612
k 100 xk [5.95747952e-10 9.07160128e-02] f(xk) 0.08229394969702622
Gradient [-1.19149590e-09 -1.81432026e+00] Step size
0.0984770902183612
k 101 xk [ 4.78412902e-10 -8.79529667e-02] f(xk) 0.07735724349398138
Gradient [-9.56825804e-10  1.75905933e+00] Step size
0.0984770902183612
k 102 xk [3.84187481e-10 8.52740780e-02] f(xk) 0.07271668382692013
Gradient [-7.68374962e-10 -1.70548156e+00] Step size
0.0984770902183612
k 103 xk [ 3.08520151e-10 -8.26767835e-02] f(xk) 0.06835450525322385
Gradient [-6.17040301e-10  1.65353567e+00] Step size
0.0984770902183612
k 104 xk [2.47755817e-10 8.01585978e-02] f(xk) 0.06425400805589657
Gradient [-4.95511634e-10 -1.60317196e+00] Step size
0.0984770902183612
k 105 xk [ 1.98959273e-10 -7.77171116e-02] f(xk) 0.06039949431208121
Gradient [-3.97918547e-10  1.55434223e+00] Step size
0.0984770902183612
k 106 xk [1.59773413e-10 7.53499886e-02] f(xk) 0.05677620779674222
Gradient [-3.19546825e-10 -1.50699977e+00] Step size
0.0984770902183612
k 107 xk [ 1.28305371e-10 -7.30549639e-02] f(xk) 0.05337027749144702
Gradient [-2.56610742e-10  1.46109928e+00] Step size
0.0984770902183612
k 108 xk [1.03035092e-10 7.08298415e-02] f(xk) 0.05016866448198212
Gradient [-2.06070184e-10 -1.41659683e+00] Step size
0.0984770902183612
k 109 xk [ 8.27418998e-11 -6.86724923e-02] f(xk) 0.047159112041511214
Gradient [-1.65483800e-10  1.37344985e+00] Step size
0.0984770902183612
k 110 xk [6.64455368e-11 6.65808521e-02] f(xk) 0.04433009870817951
Gradient [-1.32891074e-10 -1.33161704e+00] Step size
0.0984770902183612
```

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k 111 xk [ 5.33588105e-11 -6.45529195e-02] f(xk) 0.04167079417753104
Gradient [-1.06717621e-10  1.29105839e+00] Step size
0.0984770902183612
k 112 xk [4.28495697e-11 6.25867541e-02] f(xk) 0.03917101784088188
Gradient [-8.56991395e-11 -1.25173508e+00] Step size
0.0984770902183612
k 113 xk [ 3.44101678e-11 -6.06804745e-02] f(xk) 0.036821199810922266
Gradient [-6.88203357e-11  1.21360949e+00] Step size
0.0984770902183612
k 114 xk [2.76329414e-11 5.88322567e-02] f(xk) 0.034612344285341594
Gradient [-5.52658829e-11 -1.17664513e+00] Step size
0.0984770902183612
k 115 xk [ 2.21905181e-11 -5.70403323e-02] f(xk) 0.032535995108221666
Gradient [-4.43810362e-11  1.14080665e+00] Step size
0.0984770902183612
k 116 xk [1.78200028e-11 5.53029867e-02] f(xk) 0.0305842033973567
Gradient [-3.56400056e-11 -1.10605973e+00] Step size
0.0984770902183612
k 117 xk [ 1.43102787e-11 -5.36185575e-02] f(xk) 0.028749497113568126
Gradient [-2.86205575e-11  1.07237115e+00] Step size
0.0984770902183612
k 118 xk [1.14918095e-11 5.19854330e-02] f(xk) 0.027024852455516182
Gradient [-2.29836191e-11 -1.03970866e+00] Step size
0.0984770902183612
k 119 xk [ 9.22844960e-12 -5.04020505e-02] f(xk) 0.025403666970499415
Gradient [-1.84568992e-11  1.00804101e+00] Step size
0.0984770902183612
k 120 xk [7.41086787e-12 4.88668950e-02] f(xk) 0.023879734278301968
Gradient [-1.48217357e-11 -9.77337900e-01] Step size
0.0984770902183612
k 121 xk [ 5.95126646e-12 -4.73784976e-02] f(xk) 0.022447220311324206
Gradient [-1.19025329e-11  9.47569951e-01] Step size
0.0984770902183612
k 122 xk [4.77913965e-12 4.59354340e-02] f(xk) 0.021100640980036712
Gradient [-9.55827931e-12 -9.18708680e-01] Step size
0.0984770902183612
k 123 xk [ 3.83786812e-12 -4.45363236e-02] f(xk) 0.019834841178254528
Gradient [-7.67573624e-12  8.90726472e-01] Step size
0.0984770902183612
k 124 xk [3.08198395e-12 4.31798275e-02] f(xk) 0.018644975047857375
Gradient [-6.1639679e-12 -8.6359655e-01] Step size 0.0984770902183612
k 125 xk [ 2.47497433e-12 -4.18646479e-02] f(xk) 0.017526487427403547
Gradient [-4.94994866e-12  8.37292958e-01] Step size
0.0984770902183612
k 126 xk [1.98751779e-12 4.05895263e-02] f(xk) 0.016475096413617062
Gradient [-3.97503557e-12 -8.11790525e-01] Step size
0.0984770902183612
k 127 xk [ 1.59606785e-12 -3.93532425e-02] f(xk) 0.015486776968988387
Gradient [-3.1921357e-12  7.8706485e-01] Step size 0.0984770902183612
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k 128 xk [1.28171561e-12 3.81546138e-02] f(xk) 0.01455774551273371
Gradient [-2.56343123e-12 -7.63092275e-01] Step size
0.0984770902183612
k 129 xk [ 1.02927637e-12 -3.69924931e-02] f(xk) 0.013684445436122399
Gradient [-2.05855273e-12  7.39849861e-01] Step size
0.0984770902183612
k 130 xk [8.26556083e-13 3.58657685e-02] f(xk) 0.012863533486720901
Gradient [-1.65311217e-12 -7.17315370e-01] Step size
0.0984770902183612
k 131 xk [ 6.63762407e-13 -3.47733619e-02] f(xk) 0.012091866969428149
Gradient [-1.32752481e-12  6.95467238e-01] Step size
0.0984770902183612
k 132 xk [5.33031626e-13 3.37142280e-02] f(xk) 0.01136649171530389
Gradient [-1.06606325e-12 -6.74284561e-01] Step size
0.0984770902183612
k 133 xk [ 4.28048819e-13 -3.26873535e-02] f(xk) 0.010684630772131462
Gradient [-8.56097638e-13  6.53747070e-01] Step size
0.0984770902183612
k 134 xk [3.43742815e-13 3.16917557e-02] f(xk) 0.010043673773418693
Gradient [-6.87485630e-13 -6.33835113e-01] Step size
0.0984770902183612
k 135 xk [ 2.7604123e-13 -3.0726482e-02] f(xk) 0.00944116694513862
Gradient [-5.52082461e-13  6.14529639e-01] Step size
0.0984770902183612
k 136 xk [2.21673756e-13 2.97906088e-02] f(xk) 0.008874803711952694
Gradient [-4.43347512e-13 -5.95812176e-01] Step size
0.0984770902183612
k 137 xk [ 1.78014183e-13 -2.88832406e-02] f(xk) 0.008342415866954346
Gradient [-3.56028366e-13  5.77664812e-01] Step size
0.0984770902183612
k 138 xk [1.42953546e-13 2.80035092e-02] f(xk) 0.007841965271128081
Gradient [-2.85907091e-13 -5.60070184e-01] Step size
0.0984770902183612
k 139 xk [ 1.14798247e-13 -2.71505728e-02] f(xk) 0.007371536050747139
Gradient [-2.29596494e-13  5.43011457e-01] Step size
0.0984770902183612
k 140 xk [9.21882525e-14 2.63236154e-02] f(xk) 0.0069293272628390395
Gradient [-1.84376505e-13 -5.26472307e-01] Step size
0.0984770902183612
k 141 xk [ 7.40313908e-14 -2.55218455e-02] f(xk) 0.006513646000640229
Gradient [-1.48062782e-13  5.10436911e-01] Step size
0.0984770902183612
k 142 xk [5.94505989e-14 2.47444962e-02] f(xk) 0.006122900912645493
Gradient [-1.18901198e-13 -4.94889924e-01] Step size
0.0984770902183612
k 143 xk [ 4.77415549e-14 -2.39908235e-02] f(xk) 0.005755596110441083
Gradient [-9.54831098e-14  4.79816470e-01] Step size
0.0984770902183612
k 144 xk [3.83386561e-14 2.32601063e-02] f(xk) 0.0054103254419989545
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Gradient [-7.66773122e-14 -4.65202126e-01] Step size
0.0984770902183612
k 145 xk [ 3.07876975e-14 -2.25516454e-02] f(xk) 0.005085767108508579
Gradient [-6.15753950e-14  4.51032908e-01] Step size
0.0984770902183612
k 146 xk [2.47239318e-14 2.18647630e-02] f(xk) 0.004780678604137972
Gradient [-4.94478635e-14 -4.37295260e-01] Step size
0.0984770902183612
k 147 xk [ 1.98544501e-14 -2.11988018e-02] f(xk) 0.004493891959351808
Gradient [-3.97089001e-14  4.23976035e-01] Step size
0.0984770902183612
k 148 xk [1.59440331e-14 2.05531245e-02] f(xk) 0.004224309269576656
Gradient [-3.18880662e-14 -4.11062490e-01] Step size
0.0984770902183612
k 149 xk [ 1.28037891e-14 -1.99271134e-02] f(xk) 0.00397089849209574
Gradient [-2.56075783e-14  3.98542268e-01] Step size
0.0984770902183612
k 150 xk [1.02820293e-14 1.93201695e-02] f(xk) 0.0037326894950824516
Gradient [-2.05640587e-14 -3.86403390e-01] Step size
0.0984770902183612
k 151 xk [ 8.25694068e-15 -1.87317120e-02] f(xk) 0.0035087703436471928
Gradient [-1.65138814e-14  3.74634240e-01] Step size
0.0984770902183612
k 152 xk [6.63070170e-15 1.81611778e-02] f(xk) 0.0032982838086793773
Gradient [-1.32614034e-14 -3.63223557e-01] Step size
0.0984770902183612
k 153 xk [ 5.32475728e-15 -1.76080211e-02] f(xk) 0.0031004240851194294
Gradient [-1.06495146e-14  3.52160423e-01] Step size
0.0984770902183612
k 154 xk [4.27602407e-15 1.70717126e-02] f(xk) 0.0029144337070973647
Gradient [-8.55204815e-15 -3.41434252e-01] Step size
0.0984770902183612
k 155 xk [ 3.43384326e-15 -1.65517390e-02] f(xk) 0.0027396006481281397
Gradient [-6.86768651e-15  3.31034781e-01] Step size
0.0984770902183612
k 156 xk [2.75753347e-15 1.60476029e-02] f(xk) 0.0025752555952625036
Gradient [-5.51506694e-15 -3.20952058e-01] Step size
0.0984770902183612
k 157 xk [ 2.21442573e-15 -1.55588219e-02] f(xk) 0.002420769386757946
Gradient [-4.42885145e-15  3.11176438e-01] Step size
0.0984770902183612
k 158 xk [1.77828532e-15 1.50849283e-02] f(xk) 0.0022755506034604317
Gradient [-3.55657065e-15 -3.01698565e-01] Step size
0.0984770902183612
k 159 xk [ 1.42804459e-15 -1.46254686e-02] f(xk) 0.002139043304675969
Gradient [-2.85608919e-15  2.92509371e-01] Step size
0.0984770902183612
k 160 xk [1.14678524e-15 1.41800032e-02] f(xk) 0.002010724899864285
Gradient [-2.29357048e-15 -2.83600063e-01] Step size
```

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0.0984770902183612
k 161 xk [ 9.20921094e-16 -1.37481059e-02] f(xk) 0.0018901041480068076
Gradient [-1.84184219e-15  2.74962117e-01] Step size
0.0984770902183612
k 162 xk [7.39541835e-16 1.33293634e-02] f(xk) 0.0017767192769899392
Gradient [-1.47908367e-15 -2.66587267e-01] Step size
0.0984770902183612
k 163 xk [ 5.93885979e-16 -1.29233750e-02] f(xk) 0.0016701362158040626
Gradient [-1.18777196e-15  2.58467500e-01] Step size
0.0984770902183612
k 164 xk [4.76917653e-16 1.25297523e-02] f(xk) 0.0015699469327906152
Gradient [-9.53835305e-16 -2.50595046e-01] Step size
0.0984770902183612
k 165 xk [ 3.82986727e-16 -1.21481187e-02] f(xk) 0.00147576787357554
Gradient [-7.65973454e-16  2.42962374e-01] Step size
0.0984770902183612
k 166 xk [3.07555890e-16 1.17781089e-02] f(xk) 0.0013872384927090648
Gradient [-6.15111780e-16 -2.35562178e-01] Step size
0.0984770902183612
k 167 xk [ 2.46981472e-16 -1.14193690e-02] f(xk) 0.001304019873390482
Gradient [-4.93962944e-16  2.28387379e-01] Step size
0.0984770902183612
k 168 xk [1.98337439e-16 1.10715556e-02] f(xk) 0.0012257934299938398
Gradient [-3.96674877e-16 -2.21431112e-01] Step size
0.0984770902183612
k 169 xk [ 1.59274051e-16 -1.07343360e-02] f(xk) 0.001152259688427406
Gradient [-3.18548102e-16  2.14686720e-01] Step size
0.0984770902183612
k 170 xk [1.27904361e-16 1.04073875e-02] f(xk) 0.0010831371396577774
Gradient [-2.55808721e-16 -2.08147749e-01] Step size
0.0984770902183612
k 171 xk [ 1.02713062e-16 -1.00903972e-02] f(xk) 0.0010181611620095688
Gradient [-2.05426124e-16  2.01807945e-01] Step size
0.0984770902183612
k 172 xk [8.24832952e-17 9.78306193e-03] f(xk) 0.0009570830081149379
Gradient [-1.64966590e-16 -1.95661239e-01] Step size
0.0984770902183612
k 173 xk [ 6.62378654e-17 -9.48508752e-03] f(xk) 0.000899668852634677
Gradient [-1.32475731e-16  1.89701750e-01] Step size
0.0984770902183612
k 174 xk [5.31920409e-17 9.19618887e-03] f(xk) 0.0008456988971052696
Gradient [-1.06384082e-16 -1.83923777e-01] Step size
0.0984770902183612
k 175 xk [ 4.27156461e-17 -8.91608955e-03] f(xk) 0.0007949665284849962
Gradient [-8.54312922e-17  1.78321791e-01] Step size
0.0984770902183612
k 176 xk [3.43026210e-17 8.64452155e-03] f(xk) 0.0007472775281777628
Gradient [-6.86052420e-17 -1.72890431e-01] Step size
0.0984770902183612
k 177 xk [ 2.75465764e-17 -8.38122502e-03] f(xk) 0.0007024493285065482
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Gradient [-5.50931528e-17  1.67624500e-01] Step size
0.0984770902183612
k 178 xk [2.21211630e-17 8.12594803e-03] f(xk) 0.0006603103137900355
Gradient [-4.42423261e-17 -1.62518961e-01] Step size
0.0984770902183612
k 179 xk [ 1.77643075e-17 -7.87844631e-03] f(xk) 0.0006206991633467421
Gradient [-3.55286150e-17  1.57568926e-01] Step size
0.0984770902183612
k 180 xk [1.42655529e-17 7.63848306e-03] f(xk) 0.0005834642339114703
Gradient [-2.85311057e-17 -1.52769661e-01] Step size
0.0984770902183612
k 181 xk [ 1.14558926e-17 -7.40582864e-03] f(xk) 0.0005484629790997859
Gradient [-2.29117852e-17  1.48116573e-01] Step size
0.0984770902183612
k 182 xk [9.19960666e-18 7.18026047e-03] f(xk) 0.0005155614036980623
Gradient [-1.83992133e-17 -1.43605209e-01] Step size
0.0984770902183612
k 183 xk [ 7.38770567e-18 -6.96156269e-03] f(xk)
0.00048463355068994866
Gradient [-1.47754113e-17  1.39231254e-01] Step size
0.0984770902183612
k 184 xk [5.93266615e-18 6.74952605e-03] f(xk) 0.00045556101905544906
Gradient [-1.18653323e-17 -1.34990521e-01] Step size
0.0984770902183612
k 185 xk [ 4.76420275e-18 -6.54394767e-03] f(xk) 0.0004282325104966027
Gradient [-9.52840551e-18  1.30878953e-01] Step size
0.0984770902183612
k 186 xk [3.82587310e-18 6.34463083e-03] f(xk) 0.00040254340335449596
Gradient [-7.65174621e-18 -1.26892617e-01] Step size
0.0984770902183612
k 187 xk [ 3.07235140e-18 -6.15138482e-03] f(xk)
0.00037839535208643597
Gradient [-6.14470281e-18  1.23027696e-01] Step size
0.0984770902183612
k 188 xk [2.46723895e-18 5.96402474e-03] f(xk) 0.00035569591076996276
Gradient [-4.93447790e-18 -1.19280495e-01] Step size
0.0984770902183612
k 189 xk [ 1.98130592e-18 -5.78237131e-03] f(xk)
0.00033435817919236156
Gradient [-3.96261185e-18  1.15647426e-01] Step size
0.0984770902183612
k 190 xk [1.59107944e-18 5.60625071e-03] f(xk) 0.0003143004701708032
Gradient [-3.18215888e-18 -1.12125014e-01] Step size
0.0984770902183612
k 191 xk [ 1.27770969e-18 -5.43549443e-03] f(xk) 0.0002954459968295124
Gradient [-2.55541939e-18  1.08709889e-01] Step size
0.0984770902183612
k 192 xk [1.02605943e-18 5.26993908e-03] f(xk) 0.0002777225786367684
Gradient [-2.05211886e-18 -1.05398782e-01] Step size
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0.0984770902183612
k 193 xk [ 8.23972734e-19 -5.10942624e-03] f(xk)
0.00026106236507636275
Gradient [-1.64794547e-18  1.02188525e-01] Step size
0.0984770902183612
k 194 xk [6.61687859e-19 4.95380234e-03] f(xk) 0.0002454015758956412
Gradient [-1.32337572e-18 -9.90760467e-02] Step size
0.0984770902183612
k 195 xk [ 5.31365669e-19 -4.80291846e-03] f(xk) 0.0002306802569357279
Gradient [-1.06273134e-18  9.60583691e-02] Step size
0.0984770902183612
k 196 xk [4.26710979e-19 4.65663023e-03] f(xk) 0.00021684205060917304
Gradient [-8.53421959e-19 -9.31326045e-02] Step size
0.0984770902183612
k 197 xk [ 3.42668468e-19 -4.51479767e-03] f(xk)
0.00020383398014634597
Gradient [-6.85336936e-19  9.02959534e-02] Step size
0.0984770902183612
k 198 xk [2.75178481e-19 4.37728508e-03] f(xk) 0.00019160624678460467
Gradient [-5.50356962e-19 -8.75457016e-02] Step size
0.0984770902183612
k 199 xk [ 2.20980929e-19 -4.24396088e-03] f(xk)
0.00018011203912382108
Gradient [-4.41961857e-19  8.48792175e-02] Step size
0.0984770902183612
k 200 xk [1.77457811e-19 4.11469748e-03] f(xk) 0.00016930735391841814
Gradient [-3.54915622e-19 -8.22939497e-02] Step size
0.0984770902183612
k 201 xk [ 1.42506753e-19 -3.98937122e-03] f(xk)
0.00015915082761985892
Gradient [-2.85013507e-19  7.97874245e-02] Step size
0.0984770902183612
k 202 xk [1.14439452e-19 3.86786217e-03] f(xk) 0.0001496035780246792
Gradient [-2.28878905e-19 -7.73572435e-02] Step size
0.0984770902183612
k 203 xk [ 9.19001239e-20 -3.75005407e-03] f(xk)
0.00014062905542184894
Gradient [-1.83800248e-19  7.50010814e-02] Step size
0.0984770902183612
k 204 xk [7.38000103e-20 3.63583419e-03] f(xk) 0.0001321929026696076
Gradient [-1.47600021e-19 -7.27166838e-02] Step size
0.0984770902183612
k 205 xk [ 5.92647898e-20 -3.52509324e-03] f(xk)
0.00012426282366610666
Gradient [-1.18529580e-19  7.05018648e-02] Step size
0.0984770902183612
k 206 xk [4.75923417e-20 3.41772526e-03] f(xk) 0.000116808459710326
Gradient [-9.51846834e-20 -6.83545053e-02] Step size
0.0984770902183612
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k 207 xk [ 3.82188310e-20 -3.31362752e-03] f(xk)
0.00010980127327993744
Gradient [-7.64376621e-20  6.62725504e-02] Step size
0.0984770902183612
k 208 xk [3.06914725e-20 3.21270040e-03] f(xk) 0.00010321443878118114
Gradient [-6.13829450e-20 -6.42540081e-02] Step size
0.0984770902183612
k 209 xk [ 2.46466587e-20 -3.11484735e-03] f(xk) 9.70227398525142e-05
Gradient [-4.92933173e-20  6.22969469e-02] Step size
0.0984770902183612
k 210 xk [1.97923962e-20 3.01997472e-03] f(xk) 9.120247282887883e-05
Gradient [-3.95847924e-20 -6.03994943e-02] Step size
0.0984770902183612
k 211 xk [ 1.58942010e-20 -2.92799173e-03] f(xk) 8.57313559970223e-05
Gradient [-3.17884021e-20  5.85598347e-02] Step size
0.0984770902183612
k 212 xk [1.27637717e-20 2.83881039e-03] f(xk) 8.058844429447169e-05
Gradient [-2.55275434e-20 -5.67762078e-02] Step size
0.0984770902183612
k 213 xk [ 1.02498935e-20 -2.75234535e-03] f(xk) 7.575404912560513e-05
Gradient [-2.0499787e-20  5.5046907e-02] Step size 0.0984770902183612
k 214 xk [8.23113413e-21 2.66851387e-03] f(xk) 7.12096629878518e-05
Gradient [-1.64622683e-20 -5.33702775e-02] Step size
0.0984770902183612
k 215 xk [ 6.60997785e-21 -2.58723576e-03] f(xk) 6.693788861946753e-05
Gradient [-1.32199557e-20  5.17447151e-02] Step size
0.0984770902183612
k 216 xk [5.30811508e-21 2.50843322e-03] f(xk) 6.292237239764264e-05
Gradient [-1.06162302e-20 -5.01686645e-02] Step size
0.0984770902183612
k 217 xk [ 4.26265963e-21 -2.43203087e-03] f(xk) 5.914774173196973e-05
Gradient [-8.52531925e-21  4.86406175e-02] Step size
0.0984770902183612
k 218 xk [3.42311099e-21 2.35795560e-03] f(xk) 5.559954621359544e-05
Gradient [-6.84622199e-21 -4.71591120e-02] Step size
0.0984770902183612
k 219 xk [ 2.74891497e-21 -2.28613653e-03] f(xk) 5.226420229475743e-05
Gradient [-5.49782995e-21  4.57227306e-02] Step size
0.0984770902183612
k 220 xk [2.20750468e-21 2.21650494e-03] f(xk) 4.912894128692366e-05
Gradient [-4.41500936e-21 -4.43300987e-02] Step size
0.0984770902183612
k 221 xk [ 1.77272740e-21 -2.14899419e-03] f(xk) 4.6181760478454756e-
05
Gradient [-3.54545481e-21  4.29798839e-02] Step size
0.0984770902183612
k 222 xk [1.42358133e-21 2.08353971e-03] f(xk) 4.3411377184654e-05
Gradient [-2.84716266e-21 -4.16707942e-02] Step size
0.0984770902183612
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k 223 xk [ 1.14320104e-21 -2.02007885e-03] f(xk) 4.080718555429475e-05
Gradient [-2.28640207e-21  4.04015770e-02] Step size
0.0984770902183612
k 224 xk [9.18042813e-22 1.95855089e-03] f(xk) 3.8359215967267293e-05
Gradient [-1.83608563e-21 -3.91710178e-02] Step size
0.0984770902183612
k 225 xk [ 7.37230443e-22 -1.89889697e-03] f(xk) 3.6058096867907954e-
05
Gradient [-1.47446089e-21  3.79779393e-02] Step size
0.0984770902183612
k 226 xk [5.92029825e-22 1.84105999e-03] f(xk) 3.389501888789668e-05
Gradient [-1.18405965e-21 -3.68211998e-02] Step size
0.0984770902183612
k 227 xk [ 4.75427076e-22 -1.78498463e-03] f(xk) 3.1861701121375034e-
05
Gradient [-9.50854153e-22  3.56996925e-02] Step size
0.0984770902183612
k 228 xk [3.81789726e-22 1.73061721e-03] f(xk) 2.9950359423175605e-05
Gradient [-7.63579452e-22 -3.46123443e-02] Step size
0.0984770902183612
k 229 xk [ 3.06594644e-22 -1.67790574e-03] f(xk) 2.8153676608799082e-
05
Gradient [-6.13189287e-22  3.35581147e-02] Step size
0.0984770902183612
k 230 xk [2.46209547e-22 1.62679976e-03] f(xk) 2.6464774442055734e-05
Gradient [-4.92419094e-22 -3.25359951e-02] Step size
0.0984770902183612
k 231 xk [ 1.97717547e-22 -1.57725037e-03] f(xk) 2.4877187303131476e-
05
Gradient [-3.95435095e-22  3.15450074e-02] Step size
0.0984770902183612
k 232 xk [1.58776250e-22 1.52921017e-03] f(xk) 2.338483743627224e-05
Gradient [-3.17552500e-22 -3.05842034e-02] Step size
0.0984770902183612
k 233 xk [ 1.27504604e-22 -1.48263319e-03] f(xk) 2.1982011682327256e-
05
Gradient [-2.55009207e-22  2.96526637e-02] Step size
0.0984770902183612
k 234 xk [1.02392039e-22 1.43747486e-03] f(xk) 2.0663339607076617e-05
Gradient [-2.04784078e-22 -2.87494971e-02] Step size
0.0984770902183612
k 235 xk [ 8.22254988e-23 -1.39369197e-03] f(xk) 1.9423772941611733e-
05
Gradient [-1.64450998e-22  2.78738393e-02] Step size
0.0984770902183612
k 236 xk [6.60308431e-23 1.35124262e-03] f(xk) 1.8258566256060522e-05
Gradient [-1.32061686e-22 -2.70248525e-02] Step size
0.0984770902183612
k 237 xk [ 5.30257925e-23 -1.31008621e-03] f(xk) 1.7163258792670455e-
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05
Gradient [-1.06051585e-22  2.62017242e-02] Step size
0.0984770902183612
k 238 xk [4.25821410e-23 1.27018335e-03] f(xk) 1.613365738870112e-05
Gradient [-8.5164282e-23 -2.5403667e-02] Step size 0.0984770902183612
k 239 xk [ 3.41954103e-23 -1.23149586e-03] f(xk) 1.5165820423750107e-
05
Gradient [-6.83908206e-23  2.46299171e-02] Step size
0.0984770902183612
k 240 xk [2.74604813e-23 1.19398671e-03] f(xk) 1.4256042730057797e-05
Gradient [-5.49209626e-23 -2.38797343e-02] Step size
0.0984770902183612
k 241 xk [ 2.20520247e-23 -1.15762003e-03] f(xk) 1.340084140802316e-05
Gradient [-4.41040494e-23  2.31524007e-02] Step size
0.0984770902183612
k 242 xk [1.77087863e-23 1.12236102e-03] f(xk) 1.259694249262818e-05
Gradient [-3.54175725e-23 -2.24472203e-02] Step size
0.0984770902183612
k 243 xk [ 1.42209668e-23 -1.08817592e-03] f(xk) 1.1841268419726029e-
05
Gradient [-2.84419336e-23  2.17635185e-02] Step size
0.0984770902183612
k 244 xk [1.14200879e-23 1.05503205e-03] f(xk) 1.1130926244210144e-05
Gradient [-2.28401758e-23 -2.11006410e-02] Step size
0.0984770902183612
k 245 xk [ 9.17085386e-24 -1.02289768e-03] f(xk) 1.0463196564959948e-
05
Gradient [-1.83417077e-23  2.04579535e-02] Step size
0.0984770902183612
k 246 xk [7.36461586e-24 9.91742059e-04] f(xk) 9.835523114164548e-06
Gradient [-1.47292317e-23 -1.98348412e-02] Step size
0.0984770902183612
k 247 xk [ 5.91412398e-24 -9.61535385e-04] f(xk) 9.245502971169253e-06
Gradient [-1.18282480e-23  1.92307077e-02] Step size
0.0984770902183612
k 248 xk [4.74931254e-24 9.32248752e-04] f(xk) 8.690877363380616e-06
Gradient [-9.49862507e-24 -1.86449750e-02] Step size
0.0984770902183612
k 249 xk [ 3.81391558e-24 -9.03854138e-04] f(xk) 8.169523019013142e-06
Gradient [-7.62783116e-24  1.80770828e-02] Step size
0.0984770902183612
k 250 xk [3.06274896e-24 8.76324371e-04] f(xk) 7.679444038573377e-06
Gradient [-6.12549792e-24 -1.75264874e-02] Step size
0.0984770902183612
k 251 xk [ 2.45952775e-24 -8.49633112e-04] f(xk) 7.218764253963027e-06
Gradient [-4.91905550e-24  1.69926622e-02] Step size
0.0984770902183612
k 252 xk [1.97511348e-24 8.23754821e-04] f(xk) 6.785720045949451e-06
Gradient [-3.95022695e-24 -1.64750964e-02] Step size
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0.0984770902183612
k 253 xk [ 1.58610662e-24 -7.98664735e-04] f(xk) 6.378653592506702e-06
Gradient [-3.17221324e-24  1.59732947e-02] Step size
0.0984770902183612
k 254 xk [1.27371629e-24 7.74338848e-04] f(xk) 5.9960065221797315e-06
Gradient [-2.54743258e-24 -1.54867770e-02] Step size
0.0984770902183612
k 255 xk [ 1.02285254e-24 -7.50753884e-04] f(xk) 5.6363139481749664e-
06
Gradient [-2.04570509e-24  1.50150777e-02] Step size
0.0984770902183612
k 256 xk [8.21397459e-25 7.27887276e-04] f(xk) 5.298198860337969e-06
Gradient [-1.64279492e-24 -1.45577455e-02] Step size
0.0984770902183612
k 257 xk [ 6.59619796e-25 -7.05717143e-04] f(xk) 4.98036685354901e-06
Gradient [-1.31923959e-24  1.41143429e-02] Step size
0.0984770902183612
k 258 xk [5.29704919e-25 6.84222272e-04] f(xk) 4.681601172355284e-06
Gradient [-1.05940984e-24 -1.36844454e-02] Step size
0.0984770902183612
k 259 xk [ 4.25377321e-25 -6.63382096e-04] f(xk) 4.400758052869142e-06
Gradient [-8.50754642e-25  1.32676419e-02] Step size
0.0984770902183612
k 260 xk [3.41597479e-25 6.43176674e-04] f(xk) 4.1367623440997565e-06
Gradient [-6.83194959e-25 -1.28635335e-02] Step size
0.0984770902183612
k 261 xk [ 2.74318428e-25 -6.23586673e-04] f(xk) 3.8886033919553384e-
06
Gradient [-5.48636856e-25  1.24717335e-02] Step size
0.0984770902183612
k 262 xk [2.20290267e-25 6.04593348e-04] f(xk) 3.6553311701586863e-06
Gradient [-4.40580533e-25 -1.20918670e-02] Step size
0.0984770902183612
k 263 xk [ 1.76903178e-25 -5.86178526e-04] f(xk) 3.4360526432640467e-
06
Gradient [-3.53806355e-25  1.17235705e-02] Step size
0.0984770902183612
k 264 xk [1.42061357e-25 5.68324586e-04] f(xk) 3.2299283478518032e-06
Gradient [-2.84122715e-25 -1.13664917e-02] Step size
0.0984770902183612
k 265 xk [ 1.14081779e-25 -5.51014444e-04] f(xk) 3.0361691788128353e-
06
Gradient [-2.28163558e-25  1.10202889e-02] Step size
0.0984770902183612
k 266 xk [9.16128958e-26 5.34231539e-04] f(xk) 2.8540333684194677e-06
Gradient [-1.83225792e-25 -1.06846308e-02] Step size
0.0984770902183612
k 267 xk [ 7.3569353e-26 -5.1795981e-04] f(xk) 2.6828236466179825e-06
Gradient [-1.47138706e-25  1.03591962e-02] Step size
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0.0984770902183612
k 268 xk [5.90795614e-26 5.02183689e-04] f(xk) 2.5218845716714676e-06
Gradient [-1.18159123e-25 -1.00436738e-02] Step size
0.0984770902183612
k 269 xk [ 4.74435948e-26 -4.86888080e-04] f(xk) 2.3706000209339126e-
06
Gradient [-9.48871896e-26  9.73776159e-03] Step size
0.0984770902183612
k 270 xk [3.80993805e-26 4.72058347e-04] f(xk) 2.22839083214946e-06
Gradient [-7.61987609e-26 -9.44116695e-03] Step size
0.0984770902183612
k 271 xk [ 3.05955482e-26 -4.57680302e-04] f(xk) 2.0947125862470386e-
06
Gradient [-6.11910964e-26  9.15360604e-03] Step size
0.0984770902183612
k 272 xk [2.45696271e-26 4.43740186e-04] f(xk) 1.9690535231422384e-06
Gradient [-4.91392542e-26 -8.87480371e-03] Step size
0.0984770902183612
k 273 xk [ 1.97305363e-26 -4.30224660e-04] f(xk) 1.8509325825674917e-
06
Gradient [-3.94610726e-26  8.60449320e-03] Step size
0.0984770902183612
k 274 xk [1.58445247e-26 4.17120793e-04] f(xk) 1.7398975624302958e-06
Gradient [-3.16890494e-26 -8.34241587e-03] Step size
0.0984770902183612
k 275 xk [ 1.27238793e-26 -4.04416047e-04] f(xk) 1.6355233876491007e-
06
Gradient [-2.54477587e-26  8.08832093e-03] Step size
0.0984770902183612
k 276 xk [1.02178581e-26 3.92098264e-04] f(xk) 1.5374104828394787e-06
Gradient [-2.04357162e-26 -7.84196527e-03] Step size
0.0984770902183612
k 277 xk [ 8.20540824e-27 -3.80155658e-04] f(xk) 1.4451832426206987e-
06
Gradient [-1.64108165e-26  7.60311316e-03] Step size
0.0984770902183612
k 278 xk [6.58931878e-27 3.68576803e-04] f(xk) 1.3584885936866242e-06
Gradient [-1.31786376e-26 -7.37153605e-03] Step size
0.0984770902183612
k 279 xk [ 5.29152490e-27 -3.57350618e-04] f(xk) 1.2769946431360802e-
06
Gradient [-1.05830498e-26  7.14701236e-03] Step size
0.0984770902183612
k 280 xk [4.24933695e-27 3.46466363e-04] f(xk) 1.2003894078881145e-06
Gradient [-8.49867390e-27 -6.92932726e-03] Step size
0.0984770902183612
k 281 xk [ 3.41241227e-27 -3.35913623e-04] f(xk) 1.1283796203179758e-
06
Gradient [-6.82482455e-27  6.71827246e-03] Step size
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0.0984770902183612
k 282 xk [2.74032341e-27 3.25682300e-04] f(xk) 1.0606896055414164e-06
Gradient [-5.48064682e-27 -6.51364600e-03] Step size
0.0984770902183612
k 283 xk [ 2.20060526e-27 -3.15762605e-04] f(xk) 9.970602260492479e-07
Gradient [-4.40121052e-27  6.31525210e-03] Step size
0.0984770902183612
k 284 xk [1.76718685e-27 3.06145046e-04] f(xk) 9.372478896518797e-07
Gradient [-3.53437371e-27 -6.12290091e-03] Step size
0.0984770902183612
k 285 xk [ 1.41913202e-27 -2.96820420e-04] f(xk) 8.810236169359683e-07
Gradient [-2.83826403e-27  5.93640840e-03] Step size
0.0984770902183612
k 286 xk [1.13962803e-27 2.87779806e-04] f(xk) 8.281721646631175e-07
Gradient [-2.27925607e-27 -5.75559611e-03] Step size
0.0984770902183612
k 287 xk [ 9.15173528e-28 -2.79014552e-04] f(xk) 7.784912017547448e-07
Gradient [-1.83034706e-27  5.58029104e-03] Step size
0.0984770902183612
k 288 xk [7.34926276e-28 2.70516272e-04] f(xk) 7.317905347085338e-07
Gradient [-1.46985255e-27 -5.41032544e-03] Step size
0.0984770902183612
k 289 xk [ 5.90179473e-28 -2.62276835e-04] f(xk) 6.878913794811399e-07
Gradient [-1.18035895e-27  5.24553669e-03] Step size
0.0984770902183612
k 290 xk [4.73941159e-28 2.54288355e-04] f(xk) 6.466256770496974e-07
Gradient [-9.47882318e-28 -5.08576711e-03] Step size
0.0984770902183612
k 291 xk [ 3.80596466e-28 -2.46543191e-04] f(xk) 6.078354500318944e-07
Gradient [-7.61192933e-28  4.93086382e-03] Step size
0.0984770902183612
k 292 xk [3.05636401e-28 2.39033930e-04] f(xk) 5.713721979015688e-07
Gradient [-6.11272802e-28 -4.78067860e-03] Step size
0.0984770902183612
k 293 xk [ 2.45440034e-28 -2.31753388e-04] f(xk) 5.370963284845249e-07
Gradient [-4.90880069e-28  4.63506776e-03] Step size
0.0984770902183612
k 294 xk [1.97099593e-28 2.24694598e-04] f(xk) 5.048766235581739e-07
Gradient [-3.94199187e-28 -4.49389196e-03] Step size
0.0984770902183612
k 295 xk [ 1.58280005e-28 -2.17850806e-04] f(xk) 4.745897365091491e-07
Gradient [-3.16560009e-28  4.35701612e-03] Step size
0.0984770902183612
k 296 xk [1.27106096e-28 2.11215463e-04] f(xk) 4.461197201257842e-07
Gradient [-2.54212192e-28 -4.22430927e-03] Step size
0.0984770902183612
k 297 xk [ 1.02072019e-28 -2.04782222e-04] f(xk) 4.1935758271770243e-
07
Gradient [-2.04144038e-28  4.09564443e-03] Step size
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0.0984770902183612
k 298 xk [8.19685082e-29 1.98544925e-04] f(xk) 3.9420087086320784e-07
Gradient [-1.63937016e-28 -3.97089849e-03] Step size
0.0984770902183612
k 299 xk [ 6.58244678e-29 -1.92497604e-04] f(xk) 3.705532771871154e-07
Gradient [-1.31648936e-28  3.84995209e-03] Step size
0.0984770902183612
k 300 xk [5.28600637e-29 1.86634475e-04] f(xk) 3.4832427166747463e-07
Gradient [-1.05720127e-28 -3.73268950e-03] Step size
0.0984770902183612
k 301 xk [ 4.24490532e-29 -1.80949925e-04] f(xk) 3.274287550597205e-07
Gradient [-8.48981064e-29  3.61899851e-03] Step size
0.0984770902183612
k 302 xk [3.40885347e-29 1.75438517e-04] f(xk) 3.0778673311145343e-07
Gradient [-6.81770694e-29 -3.50877034e-03] Step size
0.0984770902183612
k 303 xk [ 2.73746553e-29 -1.70094977e-04] f(xk) 2.893230103206499e-07
Gradient [-5.47493106e-29  3.40189953e-03] Step size
0.0984770902183612
k 304 xk [2.19831025e-29 1.64914190e-04] f(xk) 2.7196690206491534e-07
Gradient [-4.39662050e-29 -3.29828381e-03] Step size
0.0984770902183612
k 305 xk [ 1.76534386e-29 -1.59891202e-04] f(xk) 2.556519639997263e-07
Gradient [-3.53068771e-29  3.19782404e-03] Step size
0.0984770902183612
k 306 xk [1.41765200e-29 1.55021204e-04] f(xk) 2.403157376897178e-07
Gradient [-2.83530401e-29 -3.10042409e-03] Step size
0.0984770902183612
k 307 xk [ 1.13843952e-29 -1.50299538e-04] f(xk) 2.2589951149921573e-
07
Gradient [-2.27687903e-29  3.00599076e-03] Step size
0.0984770902183612
k 308 xk [9.14219093e-30 1.45721685e-04] f(xk) 2.1234809582663353e-07
Gradient [-1.82843819e-29 -2.91443371e-03] Step size
0.0984770902183612
k 309 xk [ 7.34159821e-30 -1.41283266e-04] f(xk) 1.9960961182226228e-
07
Gradient [-1.46831964e-29  2.82566532e-03] Step size
0.0984770902183612
k 310 xk [5.89563975e-30 1.36980032e-04] f(xk) 1.876352927806045e-07
Gradient [-1.17912795e-29 -2.73960065e-03] Step size
0.0984770902183612
k 311 xk [ 4.73446886e-30 -1.32807868e-04] f(xk) 1.7637929744692062e-
07
Gradient [-9.46893771e-30  2.65615736e-03] Step size
0.0984770902183612
k 312 xk [3.80199542e-30 1.28762780e-04] f(xk) 1.6579853452327196e-07
Gradient [-7.60399085e-30 -2.57525560e-03] Step size
0.0984770902183612
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k 313 xk [ 3.05317653e-30 -1.24840898e-04] f(xk) 1.5585249770221569e-07
Gradient [-6.10635306e-30  2.49681796e-03] Step size 0.0984770902183612
k 314 xk [2.45184065e-30 1.21038469e-04] f(xk) 1.4650311059661218e-07
Gradient [-4.90368130e-30 -2.42076939e-03] Step size 0.0984770902183612
k 315 xk [ 1.96894038e-30 -1.17351856e-04] f(xk) 1.377145809718906e-07
Gradient [-3.93788077e-30  2.34703712e-03] Step size 0.0984770902183612
k 316 xk [1.58114934e-30 1.13777530e-04] f(xk) 1.2945326372272934e-07
Gradient [-3.16229869e-30 -2.27555060e-03] Step size 0.0984770902183612
k 317 xk [ 1.26973537e-30 -1.10312072e-04] f(xk) 1.2168753206958587e-07
Gradient [-2.53947074e-30  2.20624144e-03] Step size 0.0984770902183612
k 318 xk [1.01965568e-30 1.06952165e-04] f(xk) 1.1438765648197815e-07
Gradient [-2.03931136e-30 -2.13904330e-03] Step size 0.0984770902183612
k 319 xk [ 8.18830233e-31 -1.03694595e-04] f(xk) 1.0752569086499974e-07
Gradient [-1.63766047e-30  2.07389191e-03] Step size 0.0984770902183612
k 320 xk [6.57558195e-31 1.00536245e-04] f(xk) 1.0107536557335673e-07
Gradient [-1.31511639e-30 -2.01072490e-03] Step size 0.0984770902183612
k 321 xk [ 5.28049360e-31 -9.74740924e-05] f(xk) 9.50119868433522e-08
Gradient [-1.05609872e-30  1.94948185e-03] Step size 0.0984770902183612
k 322 xk [4.24047831e-31 9.45052074e-05] f(xk) 8.931234225781422e-08
Gradient [-8.48095662e-31 -1.89010415e-03] Step size 0.0984770902183612
k 323 xk [ 3.40529838e-31 -9.16267493e-05] f(xk) 8.395461188205918e-08
Gradient [-6.81059676e-31  1.83253499e-03] Step size 0.0984770902183612
k 324 xk [2.73461063e-31 8.88359638e-05] f(xk) 7.891828473069193e-08
Gradient [-5.46922125e-31 -1.77671928e-03] Step size 0.0984770902183612
k 325 xk [ 2.19601763e-31 -8.61301807e-05] f(xk) 7.418408024545327e-08
Gradient [-4.39203526e-31  1.72260361e-03] Step size 0.0984770902183612
k 326 xk [1.76350278e-31 8.35068108e-05] f(xk) 6.97338744835084e-08
Gradient [-3.52700556e-31 -1.67013622e-03] Step size 0.0984770902183612
k 327 xk [ 1.41617353e-31 -8.09633440e-05] f(xk) 6.555063073360332e-08
Gradient [-2.83234707e-31  1.61926688e-03] Step size 0.0984770902183612
k 328 xk [1.13725224e-31 7.84973466e-05] f(xk) 6.161833429446698e-08
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Gradient [-2.27450447e-31 -1.56994693e-03] Step size
0.0984770902183612
k 329 xk [ 9.13265655e-32 -7.61064591e-05] f(xk) 5.7921931165771636e-
08
Gradient [-1.82653131e-31  1.52212918e-03] Step size
0.0984770902183612
k 330 xk [7.33394166e-32 7.37883937e-05] f(xk) 5.4447270416942186e-08
Gradient [-1.46678833e-31 -1.47576787e-03] Step size
0.0984770902183612
k 331 xk [ 5.88949119e-32 -7.15409323e-05] f(xk) 5.118105001318519e-08
Gradient [-1.17789824e-31  1.43081865e-03] Step size
0.0984770902183612
k 332 xk [4.72953128e-32 6.93619246e-05] f(xk) 4.811076589134325e-08
Gradient [-9.45906256e-32 -1.38723849e-03] Step size
0.0984770902183612
k 333 xk [ 3.79803032e-32 -6.72492856e-05] f(xk) 4.522466409062223e-08
Gradient [-7.59606065e-32  1.34498571e-03] Step size
0.0984770902183612
k 334 xk [3.04999237e-32 6.52009937e-05] f(xk) 4.251169575493349e-08
Gradient [-6.09998475e-32 -1.30401987e-03] Step size
0.0984770902183612
k 335 xk [ 2.44928363e-32 -6.32150890e-05] f(xk) 3.9961474834586554e-
08
Gradient [-4.89856725e-32  1.26430178e-03] Step size
0.0984770902183612
k 336 xk [1.96688698e-32 6.12896715e-05] f(xk) 3.7564238325401836e-08
Gradient [-3.93377395e-32 -1.22579343e-03] Step size
0.0984770902183612
k 337 xk [ 1.57950036e-32 -5.94228987e-05] f(xk) 3.531080889302684e-08
Gradient [-3.15900073e-32  1.18845797e-03] Step size
0.0984770902183612
k 338 xk [1.26841116e-32 5.76129844e-05] f(xk) 3.319255973937082e-08
Gradient [-2.53682233e-32 -1.15225969e-03] Step size
0.0984770902183612
k 339 xk [ 1.01859228e-32 -5.58581969e-05] f(xk) 3.120138157665578e-08
Gradient [-2.03718457e-32  1.11716394e-03] Step size
0.0984770902183612
k 340 xk [8.17976275e-33 5.41568570e-05] f(xk) 2.932965158265098e-08
Gradient [-1.63595255e-32 -1.08313714e-03] Step size
0.0984770902183612
k 341 xk [ 6.56872428e-33 -5.25073368e-05] f(xk) 2.757020421824224e-08
Gradient [-1.31374486e-32  1.05014674e-03] Step size
0.0984770902183612
k 342 xk [5.27498657e-33 5.09080581e-05] f(xk) 2.591630379561704e-08
Gradient [-1.05499731e-32 -1.01816116e-03] Step size
0.0984770902183612
k 343 xk [ 4.23605591e-33 -4.93574905e-05] f(xk) 2.4361618692048114e-
08
Gradient [-8.47211183e-33  9.87149810e-04] Step size
0.0984770902183612
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k 344 xk [3.40174699e-33 4.78541504e-05] f(xk) 2.2900197110558598e-08
Gradient [-6.80349399e-33 -9.57083008e-04] Step size
0.0984770902183612
k 345 xk [ 2.73175870e-33 -4.63965993e-05] f(xk) 2.1526444294672922e-
08
Gradient [-5.46351741e-33  9.27931987e-04] Step size
0.0984770902183612
k 346 xk [2.19372741e-33 4.49834426e-05] f(xk) 2.0235101110025025e-08
Gradient [-4.38745481e-33 -8.99668853e-04] Step size
0.0984770902183612
k 347 xk [ 1.76166362e-33 -4.36133281e-05] f(xk) 1.9021223910827846e-
08
Gradient [-3.52332725e-33  8.72266563e-04] Step size
0.0984770902183612
k 348 xk [1.41469661e-33 4.22849449e-05] f(xk) 1.7880165614126824e-08
Gradient [-2.82939322e-33 -8.45698897e-04] Step size
0.0984770902183612
k 349 xk [ 1.13606620e-33 -4.09970217e-05] f(xk) 1.680755790938424e-08
Gradient [-2.27213239e-33  8.19940435e-04] Step size
0.0984770902183612
k 350 xk [9.12313210e-34 3.97483264e-05] f(xk) 1.5799294535287235e-08
Gradient [-1.82462642e-33 -7.94966528e-04] Step size
0.0984770902183612
k 351 xk [ 7.32629310e-34 -3.85376641e-05] f(xk) 1.4851515559758213e-
08
Gradient [-1.46525862e-33  7.70753282e-04] Step size
0.0984770902183612
k 352 xk [5.88334904e-34 3.73638764e-05] f(xk) 1.3960592602986761e-08
Gradient [-1.17666981e-33 -7.47277528e-04] Step size
0.0984770902183612
k 353 xk [ 4.72459885e-34 -3.62258402e-05] f(xk) 1.3123114946912646e-
08
Gradient [-9.44919771e-34  7.24516803e-04] Step size
0.0984770902183612
k 354 xk [3.79406936e-34 3.51224664e-05] f(xk) 1.2335876477982594e-08
Gradient [-7.58813872e-34 -7.02449329e-04] Step size
0.0984770902183612
k 355 xk [ 3.04681154e-34 -3.40526995e-05] f(xk) 1.1595863413194037e-
08
Gradient [-6.09362308e-34  6.81053989e-04] Step size
0.0984770902183612
k 356 xk [2.44672927e-34 3.30155157e-05] f(xk) 1.0900242762437447e-08
Gradient [-4.89345854e-34 -6.60310314e-04] Step size
0.0984770902183612
k 357 xk [ 1.96483571e-34 -3.20099227e-05] f(xk) 1.024635148296755e-08
Gradient [-3.92967142e-34  6.40198453e-04] Step size
0.0984770902183612
k 358 xk [1.57785310e-34 3.10349582e-05] f(xk) 9.631686284483687e-09
Gradient [-3.15570621e-34 -6.20699163e-04] Step size
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0.0984770902183612
k 359 xk [ 1.26708834e-34 -3.00896893e-05] f(xk) 9.053894045789976e-09
Gradient [-2.53417668e-34  6.01793787e-04] Step size
0.0984770902183612
k 360 xk [1.01752999e-34 2.91732117e-05] f(xk) 8.510762806347504e-09
Gradient [-2.03505999e-34 -5.83464234e-04] Step size
0.0984770902183612
k 361 xk [ 8.17123207e-35 -2.82846483e-05] f(xk) 8.00021329823151e-09
Gradient [-1.63424641e-34  5.65692966e-04] Step size
0.0984770902183612
k 362 xk [6.56187376e-35 2.74231490e-05] f(xk) 7.520290986075332e-09
Gradient [-1.31237475e-34 -5.48462979e-04] Step size
0.0984770902183612
k 363 xk [ 5.26948529e-35 -2.65878893e-05] f(xk) 7.069158584527691e-09
Gradient [-1.05389706e-34  5.31757786e-04] Step size
0.0984770902183612
k 364 xk [4.23163813e-35 2.57780702e-05] f(xk) 6.645089024577933e-09
Gradient [-8.46327627e-35 -5.15561404e-04] Step size
0.0984770902183612
k 365 xk [ 3.39819931e-35 -2.49929167e-05] f(xk) 6.246458841822167e-09
Gradient [-6.79639863e-35  4.99858334e-04] Step size
0.0984770902183612
k 366 xk [2.72890975e-35 2.42316775e-05] f(xk) 5.871741961358691e-09
Gradient [-5.45781950e-35 -4.84633551e-04] Step size
0.0984770902183612
k 367 xk [ 2.19143957e-35 -2.34936244e-05] f(xk) 5.519503855519355e-09
Gradient [-4.38287914e-35  4.69872487e-04] Step size
0.0984770902183612
k 368 xk [1.75982638e-35 2.27780510e-05] f(xk) 5.1883960520709935e-09
Gradient [-3.51965277e-35 -4.55561019e-04] Step size
0.0984770902183612
k 369 xk [ 1.41322122e-35 -2.20842726e-05] f(xk) 4.8771509718626505e-
09
Gradient [-2.82644244e-35  4.41685452e-04] Step size
0.0984770902183612
k 370 xk [1.13488139e-35 2.14116255e-05] f(xk) 4.58457707615558e-09
Gradient [-2.26976279e-35 -4.28232510e-04] Step size
0.0984770902183612
k 371 xk [ 9.11361759e-36 -2.07594660e-05] f(xk) 4.309554305058567e-09
Gradient [-1.82272352e-35  4.15189321e-04] Step size
0.0984770902183612
k 372 xk [7.31865251e-36 2.01271702e-05] f(xk) 4.051029789605519e-09
Gradient [-1.46373050e-35 -4.02543403e-04] Step size
0.0984770902183612
k 373 xk [ 5.87721330e-36 -1.95141329e-05] f(xk) 3.808013821059928e-09
Gradient [-1.17544266e-35  3.90282658e-04] Step size
0.0984770902183612
k 374 xk [4.71967157e-36 1.89197676e-05] f(xk) 3.579576062015457e-09
Gradient [-9.43934314e-36 -3.78395352e-04] Step size
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0.0984770902183612
k 375 xk [ 3.79011252e-36 -1.83435056e-05] f(xk) 3.3648419847876493e-
09
Gradient [-7.58022505e-36  3.66870112e-04] Step size
0.0984770902183612
k 376 xk [3.04363402e-36 1.77847955e-05] f(xk) 3.162989523461843e-09
Gradient [-6.08726804e-36 -3.55695911e-04] Step size
0.0984770902183612
k 377 xk [ 2.44417758e-36 -1.72431028e-05] f(xk) 2.9732459267803464e-
09
Gradient [-4.88835515e-36  3.44862055e-04] Step size
0.0984770902183612
k 378 xk [1.96278658e-36 1.67179090e-05] f(xk) 2.794884799820794e-09
Gradient [-3.92557317e-36 -3.34358179e-04] Step size
0.0984770902183612
k 379 xk [ 1.57620756e-36 -1.62087116e-05] f(xk) 2.6272233231403307e-
09
Gradient [-3.15241512e-36  3.24174232e-04] Step size
0.0984770902183612
k 380 xk [1.26576689e-36 1.57150235e-05] f(xk) 2.4696196387397056e-09
Gradient [-2.53153379e-36 -3.14300470e-04] Step size
0.0984770902183612
k 381 xk [ 1.01646881e-36 -1.52363722e-05] f(xk) 2.3214703928399407e-
09
Gradient [-2.03293762e-36  3.04727445e-04] Step size
0.0984770902183612
k 382 xk [8.16271030e-37 1.47722998e-05] f(xk) 2.182208426064611e-09
Gradient [-1.63254206e-36 -2.95445997e-04] Step size
0.0984770902183612
k 383 xk [ 6.55503038e-37 -1.43223622e-05] f(xk) 2.0513006021850728e-
09
Gradient [-1.31100608e-36  2.86447245e-04] Step size
0.0984770902183612
k 384 xk [5.26398975e-37 1.38861289e-05] f(xk) 1.928245767116408e-09
Gradient [-1.05279795e-36 -2.77722579e-04] Step size
0.0984770902183612
k 385 xk [ 4.22722496e-37 -1.34631825e-05] f(xk) 1.8125728303505286e-
09
Gradient [-8.45444992e-37  2.69263650e-04] Step size
0.0984770902183612
k 386 xk [3.39465533e-37 1.30531183e-05] f(xk) 1.7038389614816078e-09
Gradient [-6.78931066e-37 -2.61062365e-04] Step size
0.0984770902183612
k 387 xk [ 2.72606377e-37 -1.26555438e-05] f(xk) 1.6016278949195701e-
09
Gradient [-5.45212755e-37  2.53110876e-04] Step size
0.0984770902183612
k 388 xk [2.18915412e-37 1.22700788e-05] f(xk) 1.5055483363016077e-09
Gradient [-4.37830823e-37 -2.45401576e-04] Step size
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0.0984770902183612
k 389 xk [ 1.75799106e-37 -1.18963543e-05] f(xk) 1.4152324645009793e-
09
Gradient [-3.51598212e-37  2.37927087e-04] Step size
0.0984770902183612
k 390 xk [1.41174737e-37 1.15340128e-05] f(xk) 1.3303345234983389e-09
Gradient [-2.82349475e-37 -2.30680257e-04] Step size
0.0984770902183612
k 391 xk [ 1.13369783e-37 -1.11827076e-05] f(xk) 1.2505294987248559e-
09
Gradient [-2.26739565e-37  2.23654153e-04] Step size
0.0984770902183612
k 392 xk [9.10411300e-38 1.08421025e-05] f(xk) 1.175511872809781e-09
Gradient [-1.82082260e-37 -2.16842051e-04] Step size
0.0984770902183612
k 393 xk [ 7.31101989e-38 -1.05118717e-05] f(xk) 1.1049944559690797e-
09
Gradient [-1.46220398e-37  2.10237433e-04] Step size
0.0984770902183612
k 394 xk [5.87108396e-38 1.01916990e-05] f(xk) 1.038707286557526e-09
Gradient [-1.17421679e-37 -2.03833980e-04] Step size
0.0984770902183612
k 395 xk [ 4.71474943e-38 -9.88127825e-06] f(xk) 9.76396597575226e-10
Gradient [-9.42949885e-38  1.97625565e-04] Step size
0.0984770902183612
k 396 xk [3.78615982e-38 9.58031234e-06] f(xk) 9.178238451720721e-10
Gradient [-7.57231964e-38 -1.91606247e-04] Step size
0.0984770902183612
k 397 xk [ 3.04045981e-38 -9.28851331e-06] f(xk) 8.627647954309323e-10
Gradient [-6.08091963e-38  1.85770266e-04] Step size
0.0984770902183612
k 398 xk [2.44162854e-38 9.00560196e-06] f(xk) 8.110086659335229e-10
Gradient [-4.88325709e-38 -1.80112039e-04] Step size
0.0984770902183612
k 399 xk [ 1.96073959e-38 -8.73130757e-06] f(xk) 7.623573188226208e-10
Gradient [-3.92147919e-38  1.74626151e-04] Step size
0.0984770902183612
k 400 xk [1.57456373e-38 8.46536770e-06] f(xk) 7.166245022714145e-10
Gradient [-3.14912747e-38 -1.69307354e-04] Step size
0.0984770902183612
k 401 xk [ 1.26444682e-38 -8.20752787e-06] f(xk) 6.736351374561169e-10
Gradient [-2.52889365e-38  1.64150557e-04] Step size
0.0984770902183612
k 402 xk [1.01540874e-38 7.95754138e-06] f(xk) 6.332246483021526e-10
Gradient [-2.03081747e-38 -1.59150828e-04] Step size
0.0984770902183612
k 403 xk [ 8.15419741e-39 -7.71516903e-06] f(xk) 5.952383314379969e-10
Gradient [-1.63083948e-38  1.54303381e-04] Step size
0.0984770902183612
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k 404 xk [6.54819414e-39 7.48017890e-06] f(xk) 5.595307639446577e-10
Gradient [-1.30963883e-38 -1.49603578e-04] Step size
0.0984770902183612
k 405 xk [ 5.25849993e-39 -7.25234615e-06] f(xk) 5.259652466334886e-10
Gradient [-1.05169999e-38  1.45046923e-04] Step size
0.0984770902183612
k 406 xk [4.22281639e-39 7.03145277e-06] f(xk) 4.944132807210373e-10
Gradient [-8.44563277e-39 -1.40629055e-04] Step size
0.0984770902183612
k 407 xk [ 3.39111505e-39 -6.81728741e-06] f(xk) 4.647540758974839e-10
Gradient [-6.78223009e-39  1.36345748e-04] Step size
0.0984770902183612
k 408 xk [2.72322076e-39 6.60964513e-06] f(xk) 4.3687408790540936e-10
Gradient [-5.44644152e-39 -1.32192903e-04] Step size
0.0984770902183612
k 409 xk [ 2.18687105e-39 -6.40832727e-06] f(xk) 4.1066658385860685e-
10
Gradient [-4.37374210e-39  1.28166545e-04] Step size
0.0984770902183612
k 410 xk [1.75615765e-39 6.21314118e-06] f(xk) 3.8603123363684815e-10
Gradient [-3.51231531e-39 -1.24262824e-04] Step size
0.0984770902183612
k 411 xk [ 1.41027506e-39 -6.02390011e-06] f(xk) 3.6287372579234463e-
10
Gradient [-2.82055012e-39  1.20478002e-04] Step size
0.0984770902183612
k 412 xk [1.13251549e-39 5.84042299e-06] f(xk) 3.4110540649747203e-10
Gradient [-2.26503099e-39 -1.16808460e-04] Step size
0.0984770902183612
k 413 xk [ 9.09461832e-40 -5.66253424e-06] f(xk) 3.2064294015155247e-
10
Gradient [-1.81892366e-39  1.13250685e-04] Step size
0.0984770902183612
k 414 xk [7.30339522e-40 5.49006366e-06] f(xk) 3.014079903473883e-10
Gradient [-1.46067904e-39 -1.09801273e-04] Step size
0.0984770902183612
k 415 xk [ 5.86496100e-40 -5.32284623e-06] f(xk) 2.8332691997619666e-
10
Gradient [-1.17299220e-39  1.06456925e-04] Step size
0.0984770902183612
k 416 xk [4.70983242e-40 5.16072194e-06] f(xk) 2.6633050932285526e-10
Gradient [-9.41966483e-40 -1.03214439e-04] Step size
0.0984770902183612
k 417 xk [ 3.78221123e-40 -5.00353566e-06] f(xk) 2.503536910722453e-10
Gradient [-7.56442247e-40  1.00070713e-04] Step size
0.0984770902183612
k 418 xk [3.03728892e-40 4.85113699e-06] f(xk) 2.3533530121221653e-10
Gradient [-6.07457784e-40 -9.70227399e-05] Step size
0.0984770902183612
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k 419 xk [ 2.43908217e-40 -4.70338011e-06] f(xk) 2.212178448795578e-10
Gradient [-4.87816434e-40  9.40676023e-05] Step size
0.0984770902183612
k 420 xk [1.95869474e-40 4.56012364e-06] f(xk) 2.0794727625255964e-10
Gradient [-3.91738948e-40 -9.12024728e-05] Step size
0.0984770902183612
k 421 xk [ 1.57292162e-40 -4.42123050e-06] f(xk) 1.954727916475342e-10
Gradient [-3.14584324e-40  8.84246101e-05] Step size
0.0984770902183612
k 422 xk [1.26312813e-40 4.28656780e-06] f(xk) 1.8374663502720446e-10
Gradient [-2.52625627e-40 -8.57313560e-05] Step size
0.0984770902183612
k 423 xk [ 1.01434977e-40 -4.15600668e-06] f(xk) 1.7272391517638902e-
10
Gradient [-2.02869953e-40  8.31201336e-05] Step size
0.0984770902183612
k 424 xk [8.14569340e-41 4.02942221e-06] f(xk) 1.6236243384507948e-10
Gradient [-1.62913868e-40 -8.05884443e-05] Step size
0.0984770902183612
k 425 xk [ 6.54136503e-41 -3.90669328e-06] f(xk) 1.5262252420098785e-
10
Gradient [-1.30827301e-40  7.81338657e-05] Step size
0.0984770902183612
k 426 xk [5.25301584e-41 3.78770246e-06] f(xk) 1.4346689897311515e-10
Gradient [-1.05060317e-40 -7.57540491e-05] Step size
0.0984770902183612
k 427 xk [ 4.21841241e-41 -3.67233587e-06] f(xk) 1.34860507704988e-10
Gradient [-8.43682483e-41  7.34467175e-05] Step size
0.0984770902183612
k 428 xk [3.38757845e-41 3.56048315e-06] f(xk) 1.2677040257108584e-10
Gradient [-6.77515691e-41 -7.12096630e-05] Step size
0.0984770902183612
k 429 xk [ 2.72038072e-41 -3.45203726e-06] f(xk) 1.191656122427662e-10
Gradient [-5.44076143e-41  6.90407451e-05] Step size
0.0984770902183612
k 430 xk [2.18459036e-41 3.34689443e-06] f(xk) 1.1201702332080614e-10
Gradient [-4.36918072e-41 -6.69378886e-05] Step size
0.0984770902183612
k 431 xk [ 1.75432616e-41 -3.24495407e-06] f(xk) 1.052972688806517e-10
Gradient [-3.50865231e-41  6.48990813e-05] Step size
0.0984770902183612
k 432 xk [1.40880429e-41 3.14611862e-06] f(xk) 9.898062370369065e-11
Gradient [-2.81760857e-41 -6.29223724e-05] Step size
0.0984770902183612
k 433 xk [ 1.13133439e-41 -3.05029352e-06] f(xk) 9.304290579346476e-11
Gradient [-2.26266879e-41  6.10058705e-05] Step size
0.0984770902183612
k 434 xk [9.08513355e-42 2.95738709e-06] f(xk) 8.746138379979486e-11
Gradient [-1.81702671e-41 -5.91477417e-05] Step size
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0.0984770902183612
k 435 xk [ 7.29577852e-42 -2.86731041e-06] f(xk) 8.221468999641148e-11
Gradient [-1.45915570e-41  5.73462082e-05] Step size
0.0984770902183612
k 436 xk [5.85884444e-42 2.77997731e-06] f(xk) 7.728273847894335e-11
Gradient [-1.17176889e-41 -5.55995462e-05] Step size
0.0984770902183612
k 437 xk [ 4.70492053e-42 -2.69530422e-06] f(xk) 7.264664827010167e-11
Gradient [-9.40984107e-42  5.39060844e-05] Step size
0.0984770902183612
k 438 xk [3.77826677e-42 2.61321011e-06] f(xk) 6.828867103768328e-11
Gradient [-7.55653353e-42 -5.22642023e-05] Step size
0.0984770902183612
k 439 xk [ 3.03412133e-42 -2.53361645e-06] f(xk) 6.419212314867605e-11
Gradient [-6.06824266e-42  5.06723290e-05] Step size
0.0984770902183612
k 440 xk [2.43653845e-42 2.45644706e-06] f(xk) 6.034132179934992e-11
Gradient [-4.87307690e-42 -4.91289413e-05] Step size
0.0984770902183612
k 441 xk [ 1.95665202e-42 -2.38162812e-06] f(xk) 5.672152497681946e-11
Gradient [-3.91330403e-42  4.76325624e-05] Step size
0.0984770902183612
k 442 xk [1.57128122e-42 2.30908802e-06] f(xk) 5.3318875022234195e-11
Gradient [-3.14256245e-42 -4.61817605e-05] Step size
0.0984770902183612
k 443 xk [ 1.26181082e-42 -2.23875737e-06] f(xk) 5.0120345579538753e-
11
Gradient [-2.52362163e-42  4.47751474e-05] Step size
0.0984770902183612
k 444 xk [1.01329190e-42 2.17056886e-06] f(xk) 4.7113691726707556e-11
Gradient [-2.02658380e-42 -4.34113772e-05] Step size
0.0984770902183612
k 445 xk [ 8.13719826e-43 -2.10445725e-06] f(xk) 4.428740309854145e-11
Gradient [-1.62743965e-42  4.20891450e-05] Step size
0.0984770902183612
k 446 xk [6.53454304e-43 2.04035928e-06] f(xk) 4.163065982156613e-11
Gradient [-1.30690861e-42 -4.08071856e-05] Step size
0.0984770902183612
k 447 xk [ 5.24753747e-43 -1.97821362e-06] f(xk) 3.913329109233865e-11
Gradient [-1.04950749e-42  3.95642723e-05] Step size
0.0984770902183612
k 448 xk [4.21401303e-43 1.91796080e-06] f(xk) 3.678573624058642e-11
Gradient [-8.42802606e-43 -3.83592160e-05] Step size
0.0984770902183612
k 449 xk [ 3.38404555e-43 -1.85954317e-06] f(xk) 3.457900812811821e-11
Gradient [-6.76809110e-43  3.71908635e-05] Step size
0.0984770902183612
k 450 xk [2.71754363e-43 1.80290484e-06] f(xk) 3.250465874338588e-11
Gradient [-5.43508726e-43 -3.60580969e-05] Step size
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0.0984770902183612
k 451 xk [ 2.18231205e-43 -1.74799161e-06] f(xk) 3.055474685998373e-11
Gradient [-4.36462411e-43  3.49598323e-05] Step size
0.0984770902183612
k 452 xk [1.75249657e-43 1.69475094e-06] f(xk) 2.8721807635271862e-11
Gradient [-3.50499314e-43 -3.38950189e-05] Step size
0.0984770902183612
k 453 xk [ 1.40733505e-43 -1.64313189e-06] f(xk) 2.699882403273819e-11
Gradient [-2.81467009e-43  3.28626378e-05] Step size
0.0984770902183612
k 454 xk [1.13015452e-43 1.59308506e-06] f(xk) 2.537919995869583e-11
Gradient [-2.26030905e-43 -3.18617011e-05] Step size
0.0984770902183612
k 455 xk [ 9.07565867e-44 -1.54456256e-06] f(xk) 2.3856735010474538e-
11
Gradient [-1.81513173e-43  3.08912512e-05] Step size
0.0984770902183612
k 456 xk [7.28816975e-44 1.49751797e-06] f(xk) 2.2425600739435132e-11
Gradient [-1.45763395e-43 -2.99503594e-05] Step size
0.0984770902183612
k 457 xk [ 5.85273425e-44 -1.45190628e-06] f(xk) 2.1080318337934638e-
11
Gradient [-1.17054685e-43  2.90381255e-05] Step size
0.0984770902183612
k 458 xk [4.70001377e-44 1.40768383e-06] f(xk) 1.9815737664821037e-11
Gradient [-9.40002755e-44 -2.81536766e-05] Step size
0.0984770902183612
k 459 xk [ 3.77432641e-44 -1.36480832e-06] f(xk) 1.8627017529161223e-
11
Gradient [-7.54865282e-44  2.72961664e-05] Step size
0.0984770902183612
k 460 xk [3.03095705e-44 1.32323872e-06] f(xk) 1.7509607156722168e-11
Gradient [-6.06191409e-44 -2.64647744e-05] Step size
0.0984770902183612
k 461 xk [ 2.43399739e-44 -1.28293526e-06] f(xk) 1.645922876825369e-11
Gradient [-4.86799477e-44  2.56587052e-05] Step size
0.0984770902183612
k 462 xk [1.95461143e-44 1.24385937e-06] f(xk) 1.5471861202877152e-11
Gradient [-3.90922285e-44 -2.48771873e-05] Step size
0.0984770902183612
k 463 xk [ 1.56964253e-44 -1.20597365e-06] f(xk) 1.4543724523885639e-
11
Gradient [-3.13928507e-44  2.41194731e-05] Step size
0.0984770902183612
k 464 xk [1.26049488e-44 1.16924187e-06] f(xk) 1.3671265548021993e-11
Gradient [-2.52098975e-44 -2.33848374e-05] Step size
0.0984770902183612
k 465 xk [ 1.01223514e-44 -1.13362887e-06] f(xk) 1.2851144242836515e-
11
```

```
Gradient [-2.02447028e-44  2.26725775e-05] Step size
0.0984770902183612
k 466 xk [8.12871198e-45 1.09910058e-06] f(xk) 1.2080220940049313e-11
Gradient [-1.62574240e-44 -2.19820117e-05] Step size
0.0984770902183612
k 467 xk [ 6.52772817e-45 -1.06562396e-06] f(xk) 1.1355544315966353e-
11
Gradient [-1.30554563e-44  2.13124793e-05] Step size
0.0984770902183612
k 468 xk [5.24206482e-45 1.03316698e-06] f(xk) 1.0674340092934541e-11
Gradient [-1.04841296e-44 -2.06633396e-05] Step size
0.0984770902183612
k 469 xk [ 4.20961824e-45 -1.00169858e-06] f(xk) 1.0034000418581737e-
11
Gradient [-8.41923648e-45  2.00339716e-05] Step size
0.0984770902183612
k 470 xk [3.38051633e-45 9.71188647e-07] f(xk) 9.432073882182215e-12
Gradient [-6.76103266e-45 -1.94237729e-05] Step size
0.0984770902183612
k 471 xk [ 2.71470951e-45 -9.41607993e-07] f(xk) 8.866256129927349e-12
Gradient [-5.42941901e-45  1.88321599e-05] Step size
0.0984770902183612
k 472 xk [2.18003612e-45 9.12928313e-07] f(xk) 8.334381043173816e-12
Gradient [-4.36007224e-45 -1.82585663e-05] Step size
0.0984770902183612
k 473 xk [ 1.75066889e-45 -8.85122164e-07] f(xk) 7.834412445897194e-12
Gradient [-3.50133779e-45  1.77024433e-05] Step size
0.0984770902183612
k 474 xk [1.40586734e-45 8.58162940e-07] f(xk) 7.364436309604513e-12
Gradient [-2.81173467e-45 -1.71632588e-05] Step size
0.0984770902183612
k 475 xk [ 1.12897589e-45 -8.32024845e-07] f(xk) 6.922653425863948e-12
Gradient [-2.25795177e-45  1.66404969e-05] Step size
0.0984770902183612
k 476 xk [9.06619366e-46 8.06682869e-07] f(xk) 6.507372518399775e-12
Gradient [-1.81323873e-45 -1.61336574e-05] Step size
0.0984770902183612
k 477 xk [ 7.28056892e-46 -7.82112765e-07] f(xk) 6.117003768383775e-12
Gradient [-1.45611378e-45  1.56422553e-05] Step size
0.0984770902183612
k 478 xk [5.84663044e-46 7.58291021e-07] f(xk) 5.7500527281359145e-12
Gradient [-1.16932609e-45 -1.51658204e-05] Step size
0.0984770902183612
k 479 xk [ 4.69511213e-46 -7.35194845e-07] f(xk) 5.405114599934137e-12
Gradient [-9.39022426e-46  1.47038969e-05] Step size
0.0984770902183612
k 480 xk [3.77039017e-46 7.12802137e-07] f(xk) 5.080868858030861e-12
Gradient [-7.54078034e-46 -1.42560427e-05] Step size
0.0984770902183612
```

```
k 481 xk [ 3.02779606e-46 -6.91091470e-07] f(xk) 4.7760741932876675e-
12
Gradient [-6.05559213e-46  1.38218294e-05] Step size
0.0984770902183612
k 482 xk [2.43145897e-46 6.70042070e-07] f(xk) 4.489563761074719e-12
Gradient [-4.86291794e-46 -1.34008414e-05] Step size
0.0984770902183612
k 483 xk [ 1.95257296e-46 -6.49633798e-07] f(xk) 4.2202407142424695e-
12
Gradient [-3.90514592e-46  1.29926760e-05] Step size
0.0984770902183612
k 484 xk [1.56800555e-46 6.29847125e-07] f(xk) 3.9670740040645525e-12
Gradient [-3.13601111e-46 -1.25969425e-05] Step size
0.0984770902183612
k 485 xk [ 1.25918031e-46 -6.10663118e-07] f(xk) 3.7290944330765894e-
12
Gradient [-2.51836061e-46  1.22132624e-05] Step size
0.0984770902183612
k 486 xk [1.01117948e-46 5.92063421e-07] f(xk) 3.5053909447000377e-12
Gradient [-2.02235896e-46 -1.18412684e-05] Step size
0.0984770902183612
k 487 xk [ 8.12023455e-47 -5.74030237e-07] f(xk) 3.295107135446644e-12
Gradient [-1.62404691e-46  1.14806047e-05] Step size
0.0984770902183612
k 488 xk [6.52092041e-47 5.56546312e-07] f(xk) 3.097437976351167e-12
Gradient [-1.30418408e-46 -1.11309262e-05] Step size
0.0984770902183612
k 489 xk [ 5.23659787e-47 -5.39594916e-07] f(xk) 2.911626731081067e-12
Gradient [-1.04731957e-46  1.07918983e-05] Step size
0.0984770902183612
k 490 xk [4.20522803e-47 5.23159828e-07] f(xk) 2.7369620589247562e-12
Gradient [-8.41045606e-47 -1.04631966e-05] Step size
0.0984770902183612
k 491 xk [ 3.37699079e-47 -5.07225324e-07] f(xk) 2.572775291567781e-12
Gradient [-6.75398158e-47  1.01445065e-05] Step size
0.0984770902183612
k 492 xk [2.71187834e-47 4.91776156e-07] f(xk) 2.41843787323164e-12
xk [2.71187834e-47 4.91776156e-07] f(xk) 2.41843787323164e-12
```

Gradient descent using a (local) optimum step size: convergent but not guaranteed faster, sometimes yes but sometimes not

```
import scipy.optimize as opt

xk = np.array([2.0,2.0])
epsilon = 0.001
print(0,xk,f(xk))

for k in range(0,100):
```

```
    dk = -gradf(xk)
    if np.linalg.norm(dk) < epsilon:
        break

    def phi(c):
        return f(xk + c * dk)

    res = opt.minimize_scalar(phi)
    ck = res.x

    print(ck)

    print('Gradient',dk,'Step size',ck)

    xk = xk + ck * dk
    print('k',k+1,'xk',xk,'f(xk)',f(xk))

print('xk',xk,'f(xk)',f(xk))
```

```
0 [2. 2.] 44.0
0.050449550449550455
Gradient [ -4. -40.] Step size 0.050449550449550455
k 1 xk [ 1.7982018  -0.01798202] f(xk) 3.236763236763237
0.459090909090909
Gradient [-3.5964036   0.35964036] Step size 0.459090909090909
k 2 xk [0.1471256 0.1471256] f(xk) 0.23810536933777465
0.050449550449550504
Gradient [-0.2942512  -2.94251203] Step size 0.050449550449550504
k 3 xk [ 0.13228076 -0.00132281] f(xk) 0.017515697862464537
0.459090909090899
Gradient [-0.26456152  0.02645615] Step size 0.459090909090899
k 4 xk [0.01082297 0.01082297] f(xk) 0.0012885037933520768
0.05044955044955062
Gradient [-0.02164594 -0.21645943] Step size 0.05044955044955062
k 5 xk [ 9.73094326e-03 -9.73094326e-05] f(xk) 9.478594792620501e-05
0.4590909090908909
Gradient [-0.01946189  0.00194619] Step size 0.4590909090908909
k 6 xk [0.00079617 0.00079617] f(xk) 6.972719809305238e-06
0.050449550449550705
Gradient [-0.00159234 -0.01592336] Step size 0.050449550449550705
k 7 xk [ 7.15835441e-04 -7.15835441e-06] f(xk) 5.129327986140909e-07
0.4590909090908847
Gradient [-0.00143167  0.00014317] Step size 0.4590909090908847
k 8 xk [5.85683542e-05 5.85683542e-05] f(xk) 3.77327733064808e-08
0.050449550449550726
Gradient [-0.00011714 -0.00117137] Step size 0.050449550449550726
k 9 xk [ 5.265886e-05 -5.265886e-07] f(xk) 2.7757284877184365e-09
xk [ 5.265886e-05 -5.265886e-07] f(xk) 2.7757284877184365e-09
```

Conjugate gradient descent: convergent and fast(er)

```python
from scipy.linalg import solve, norm
import numpy as np

# define the quadratic function f by defining the matrix A and the
vector b
A = np.array([[2.0, 0.0], [0.0, 20.0]])
b = np.array([0.0,0.0])

print("f(x) = 0.5 * x^T *",A," * x -",b,"* x + c")

def f(x):
    return 0.5 * A.dot(x).dot(x) - b.dot(x)

# implement the conjugate gradient method for quadratic functions

# choose the starting point
xk = np.array([2.0, 2.0])

k = 0
epsilon = 0.001
kmax = 100000

# compute the residum of A x = b as b - A x
rk = b - A @ xk

# iterate as long as the residuum is too large of the number of
iterations have been exceeded
while (norm(rk) >= epsilon) & (k <= kmax):
    k += 1
    print("Iteration k =",k)

    dk = rk
    #print("dk =",dk)
    z = A @ dk
    #print("z =", z)

    error = rk.dot(rk)
    #print("error", np.sqrt(error))
    alphak = error/dk.dot(z)
    #print("alphak",alphak)

    xk += alphak * dk
    print("x =",xk)

    rk -= alphak * z
    #print("rk =", rk)

    betak = rk.dot(rk)/error
    #print("betak",betak)
```

```
    dk = rk + betak * dk
    #print("dk",dk)

print("x =",xk,"value:",f(xk))
```

```
f(x) = 0.5 * x^T * [[ 2.  0.]
 [ 0. 20.]]  * x - [0. 0.] * x + c
Iteration k = 1
x = [ 1.7982018  -0.01798202]
Iteration k = 2
x = [0.1471256 0.1471256]
Iteration k = 3
x = [ 0.13228076 -0.00132281]
Iteration k = 4
x = [0.01082297 0.01082297]
Iteration k = 5
x = [ 9.73094326e-03 -9.73094326e-05]
Iteration k = 6
x = [0.00079617 0.00079617]
Iteration k = 7
x = [ 7.15835441e-04 -7.15835441e-06]
Iteration k = 8
x = [5.85683542e-05 5.85683542e-05]
Iteration k = 9
x = [ 5.265886e-05 -5.265886e-07]
x = [ 5.265886e-05 -5.265886e-07] value: 2.775728487711379e-09
```

Global Newton method using gradient descent: for quadratic functions more or less unbeatable

```
xk = np.array([2.0, 2.0])
k = 0
rho = 0.05
p = 2
sigma = 0.01
beta = 0.9
epsilon = 0.000001
kmax = 100000

print("k =", 0, ", xk =", xk, "value =", f(xk))

while (norm(gradf(xk)) >= epsilon) & (k <= kmax):
    grad = gradf(xk)
    hess = np.array([[2.0, 0.0],[0.0,20.0]])

    dk = solve(hess,-grad)

    alphak = 1

    if np.dot(grad,dk) > -rho * norm(dk)**p:
```

```
        dk = -grad

        def phi(alpha):
            return f(xk + alpha * dk)

        while phi(alphak) > phi(0) + sigma * alphak * np.dot(grad,dk):
            alphak = alphak * beta

        print("gradient step:", alphak * dk)
    else:
        print("newton step:", dk)

    k += 1
    xk = xk + alphak * dk

    print("k =", k, ", xk =", xk, "value =", f(xk))

print("Minimum at ", xk, "with a value of ", f(xk))
```

```
k = 0 , xk = [2. 2.] value = 44.0
newton step: [-2. -2.]
k = 1 , xk = [0. 0.] value = 0.0
Minimum at  [0. 0.] with a value of  0.0
```